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% 2D Lid-Driven Cavity Flow Solver
% Solves the incompressible Navier-Stokes equations using Finite Difference Method
% with a staggered grid and Chorin's projection method
% MODIFIED to loop through multiple Reynolds numbers

clear all; close all; clc;

%% Parameters
N = 41; % Number of grid points in each direction (Use N=61 or higher
for better resolution at high Re)
L = 1.0; % Length of cavity
% Re = 100; % Original: Single Reynolds number
Re_values = 100:100:1000; % MODIFIED: Range of Reynolds numbers
dt = 0.0005; % Time step (May need to be smaller for higher Re stability)
nIterations = 15000; % Number of time iterations (May need more for higher Re to
converge)
plotInterval = 1000; % Interval between intermediate plots (within a Re simulation)
eps_p = 1e-6; % Convergence tolerance for pressure Poisson equation
eps_vel = 1e-7; % Convergence tolerance for velocity field (steady state)

% Grid setup (Done once outside the Re loop)
h = L/(N-1); % Grid spacing
x = linspace(0, L, N);
y = linspace(0, L, N);
[X, Y] = meshgrid(x, y); % Note: X(i,j) = x(j), Y(i,j) = y(i)

%% Loop over Reynolds numbers
for re_idx = 1:length(Re_values)
    Re = Re_values(re_idx);
    fprintf('\n--- Starting Simulation for Re = %d ---\n', Re);

    % Adjust dt based on Re? (Optional, simple stability criteria)
    % dt_cfl = h / 1.0; % Based on lid velocity U=1
    % dt_visc = h^2 * Re / 4; % More relevant for low Re (diffusion) or explicit
diffusion
    % dt_stable = min(dt_cfl, 0.25 * h^2 * Re); % Rough estimate
    % dt = min(0.001, 0.5 * h^2 * Re); % Example adaptive dt (can be complex)
    % Using fixed dt=0.0005 for this range, might need tuning.

    % Initialize variables for EACH Re simulation
    u = zeros(N, N); % x-velocity component (u(i,j) corresponds to y(i), x(j))
    v = zeros(N, N); % y-velocity component (v(i,j) corresponds to y(i), x(j))
    p = zeros(N, N); % pressure
    psi = zeros(N, N); % stream function

    % Set boundary condition: top lid moving with u = 1
    % u is zero elsewhere on boundaries due to initialization
    u(N, 2:N-1) = 1.0; % Lid velocity (at y=L, for x from h to L-h)
    % v is zero on all boundaries

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%% Main time-stepping loop (Inner loop for the current Re)
tic; % Start timer for this Re simulation
for iter = 1:nIterations
    % Store previous velocities
    u_old = u;
    v_old = v;

    % Step 1: Compute tentative velocity field (momentum equations without
    pressure gradient)
    % Using loops as in the original code. Vectorization possible for speed.
    u_tent = u;
    v_tent = v;

    % Interior points
    for i = 2:N-1
        for j = 2:N-1
            % Compute derivatives using central differences
            u_x = (u(i,j+1) - u(i,j-1))/(2*h); % d/dx at (i,j)
            u_y = (u(i+1,j) - u(i-1,j))/(2*h); % d/dy at (i,j)
            v_x = (v(i,j+1) - v(i,j-1))/(2*h); % d/dx at (i,j)
            v_y = (v(i+1,j) - v(i-1,j))/(2*h); % d/dy at (i,j)

            u_xx = (u(i,j+1) - 2*u(i,j) + u(i,j-1))/(h^2); % d^2/dx^2
            u_yy = (u(i+1,j) - 2*u(i,j) + u(i-1,j))/(h^2); % d^2/dy^2
            v_xx = (v(i,j+1) - 2*v(i,j) + v(i,j-1))/(h^2); % d^2/dx^2
            v_yy = (v(i+1,j) - 2*v(i,j) + v(i-1,j))/(h^2); % d^2/dy^2

            % Compute tentative velocities using Euler forward step
            % Convection term: u*du/dx + v*du/dy
            conv_u = u(i,j)*u_x + v(i,j)*u_y;
            % Diffusion term: (1/Re)*(d^2u/dx^2 + d^2u/dy^2)
            diff_u = (1/Re)*(u_xx + u_yy);
            u_tent(i,j) = u(i,j) + dt * (-conv_u + diff_u);

            % Convection term: u*dv/dx + v*dv/dy
            conv_v = u(i,j)*v_x + v(i,j)*v_y;
            % Diffusion term: (1/Re)*(d^2v/dx^2 + d^2v/dy^2)
            diff_v = (1/Re)*(v_xx + v_yy);
            v_tent(i,j) = v(i,j) + dt * (-conv_v + diff_v);
        end
    end

    % Apply boundary conditions to tentative velocities
    % Note: Using a non-staggered grid, applying BCs directly on u,v nodes.
    % No-slip on walls (u=0, v=0)
    u_tent(1,:) = 0; % Bottom wall (y=0)
    v_tent(1,:) = 0;
    u_tent(:,1) = 0; % Left wall (x=0)
    v_tent(:,1) = 0;
    u_tent(:,N) = 0; % Right wall (x=L)

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v_tent(:,N) = 0;
% Moving lid (u=1, v=0) at y=L
u_tent(N,:) = 1.0;      % Top lid (y=L)
v_tent(N,:) = 0.0;
% Ensure corner consistency (optional but good practice)
u_tent(N,1) = 0; u_tent(N,N) = 0; % Overwrite corners based on side walls

% Step 2: Solve pressure Poisson equation: laplacian(p) = (1/
dt)*divergence(u_tent)
% RHS of Poisson equation using central differences for divergence
rhs = zeros(N, N);
for i = 2:N-1
    for j = 2:N-1
        div_u_tent = (u_tent(i,j+1) - u_tent(i,j-1))/(2*h) +
(v_tent(i+1,j) - v_tent(i-1,j))/(2*h);
        rhs(i,j) = (1/dt) * div_u_tent;
    end
end

% Solve Poisson equation using iterative method (Jacobi or Gauss-Seidel)
p_new = p; % Use current pressure as initial guess
max_iter_p = 1000; % Max iterations for pressure solver per time step
p_converged = false;
for k = 1:max_iter_p
    p_old_iter = p_new;

    % Update interior points using Gauss-Seidel (updates happen immediately)
    for i = 2:N-1
        for j = 2:N-1
            p_new(i,j) = 0.25 * (p_new(i,j+1) + p_new(i,j-1) + ... % Using
latest values where available
                                p_new(i+1,j) + p_new(i-1,j) - h^2 *
rhs(i,j));
        end
    end

    % Apply Neumann boundary conditions for pressure (dp/dn = 0)
    % This implies that the normal velocity component on the boundary
remains unchanged by the pressure correction
    p_new(1,:) = p_new(2,:);      % Bottom wall: dp/dy = 0 => p(1,j) =
p(2,j)
    p_new(N,:) = p_new(N-1,:);    % Top wall: dp/dy = 0 => p(N,j) =
p(N-1,j)
    p_new(:,1) = p_new(:,2);      % Left wall: dp/dx = 0 => p(i,1) = p(i,2)
    p_new(:,N) = p_new(:,N-1);    % Right wall: dp/dx = 0 => p(i,N) =
p(i,N-1)

    % Check convergence of pressure solver
    if max(max(abs(p_new - p_old_iter))) < eps_p
        p_converged = true;
    end
end

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        break;
    end
end
% if ~p_converged
%     fprintf('Warning (Re=%d, iter=%d): Pressure solver did not converge
in %d iterations. Res: %e\n', Re, iter, max_iter_p, max(max(abs(p_new -
p_old_iter))));
% end
p = p_new; % Update pressure field

% Step 3: Correct velocities with pressure gradient
% u = u_tent - dt * grad(p)_x
% v = v_tent - dt * grad(p)_y
for i = 2:N-1
    for j = 2:N-1
        p_x = (p(i,j+1) - p(i,j-1))/(2*h); % dp/dx at (i,j)
        p_y = (p(i+1,j) - p(i-1,j))/(2*h); % dp/dy at (i,j)
        u(i,j) = u_tent(i,j) - dt * p_x;
        v(i,j) = v_tent(i,j) - dt * p_y;
    end
end

% Apply boundary conditions AGAIN to enforce them strictly after correction
% No-slip on walls
u(1,:) = 0; v(1,:) = 0; % Bottom
u(:,1) = 0; v(:,1) = 0; % Left
u(:,N) = 0; v(:,N) = 0; % Right
% Moving lid
u(N,:) = 1.0; v(N,:) = 0.0; % Top
% Ensure corner consistency
u(N,1) = 0; u(N,N) = 0; % Overwrite corners based on side walls

% Check for convergence of the overall solution (steady state)
% Calculate change from previous time step
err_u = max(max(abs(u - u_old)));
err_v = max(max(abs(v - v_old)));
% Normalize by dt or use absolute change; using max absolute change here
err = max(err_u, err_v);

% Display progress periodically
if mod(iter, 500) == 0 % Report every 500 iterations
    fprintf('Re=%d, Iteration %d/%d, Max Vel Change = %e\n', Re, iter,
nIterations, err);
end

% Plot results at intervals OR if converged OR at final iteration
plot_now = mod(iter, plotInterval) == 0 || (err < eps_vel && iter > 1) ||
iter == nIterations;

if plot_now

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% Compute stream function (psi) from velocities
% Definition:  $u = d(\psi)/dy$ ,  $v = -d(\psi)/dx$ 
% Integrate  $u$  dy:  $\psi(i,j) = \text{integral}(u \text{ dy})$  from  $y=0$ 
psi = zeros(N, N);
psi(:,1) = 0; %  $\psi = 0$  on bottom wall ( $y=0$ ), assuming  $i$  is  $y$ -index
for i = 2:N % Loop over  $y$ -direction (rows)
    for j = 1:N % Loop over  $x$ -direction (columns)
        % Trapezoidal rule:  $\psi(i,j) = \psi(i-1,j) + h * (u(i,j) +$ 
u(i-1,j))/2
        % Need  $u$  at node  $(i,j)$  and  $(i-1,j)$ 
         $\psi(i,j) = \psi(i-1,j) + 0.5*(u(i,j) + u(i-1,j)) * h;$ 
    end
end

% Plot streamlines and velocity vectors
figure(100 + re_idx); % Use a unique figure number for each Re
streamlines plot
clf; % Clear the figure window
contourf(X, Y, psi, 20, 'LineColor', 'none'); %  $\psi(i,j)$  corresponds to
Y(i), X(j)
colormap(jet);
colorbar;
hold on;
% Decimate quiver plot for clarity
skip = max(1, floor(N/20)); % Adjust decimation based on N
quiver(X(1:skip:N, 1:skip:N), Y(1:skip:N, 1:skip:N), u(1:skip:N,
1:skip:N), v(1:skip:N, 1:skip:N), 1.5, 'k');
title(sprintf('Lid-driven cavity, Re = %d, Iteration = %d', Re, iter));
xlabel('x/L');
ylabel('y/L');
axis equal; axis([0 L 0 L]); % Use axis equal and set limits
hold off;
drawnow;

% Plot velocity profiles along centerlines
figure(200 + re_idx); % Use a unique figure number for each Re profile
plot

clf;
mid_N = round((N+1)/2); % Index for centerline (works for odd/even N)

% x-velocity along vertical centerline ( $x=L/2$ )
subplot(1,2,1);
plot(u(:, mid_N), y, 'b-o', 'LineWidth', 1, 'MarkerSize', 3); % Plot  $u$ 
at  $x=L/2$  vs  $y$ 
title(sprintf('u-velocity at  $x=L/2$  (Re=%d)', Re));
xlabel('u / U_{lid}');
ylabel('y/L');
grid on;
xlim([-0.5 1.1]); % Adjust limits if needed

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        % y-velocity along horizontal centerline (y=L/2)
        subplot(1,2,2);
        plot(x, v(mid_N, :), 'r-s', 'LineWidth', 1, 'MarkerSize', 3); % Plot v
at y=L/2 vs x
        title(sprintf('v-velocity at y=L/2 (Re=%d)', Re));
        xlabel('x/L');
        ylabel('v / U_{lid}');
        grid on;
        ylim([-0.6 0.6]); % Adjust limits if needed

        drawnow;
    end

    % Break INNER loop (time stepping) if converged
    if err < eps_vel && iter > 10 % Check convergence only after some initial
iterations
        fprintf('Solution for Re=%d converged after %d iterations (Max Vel
Change = %e)\n', Re, iter, err);
        break; % Exit the time-stepping loop for this Re
    end
end % End of inner time-stepping loop (iter)

simulation_time = toc;
fprintf('Simulation time for Re=%d: %.2f seconds\n', Re, simulation_time);

% Check if the loop finished due to max iterations without converging
if iter == nIterations && err >= eps_vel
    fprintf('Warning: Solution for Re=%d did not fully converge within %d
iterations. Final Max Vel Change = %e\n', Re, nIterations, err);
end

% --- Final Calculations and Plots for this Re (ensure final state is plotted
if not done already) ---
% The plotting logic inside the loop should have covered the final state either
via convergence or max iterations.

% Calculate final vorticity for this Re
omega = zeros(N, N);
for i = 2:N-1
    for j = 2:N-1
        % Central difference: omega = dv/dx - du/dy
        v_x = (v(i,j+1) - v(i,j-1))/(2*h);
        u_y = (u(i+1,j) - u(i-1,j))/(2*h);
        omega(i,j) = v_x - u_y;
    end
end

% Note: Vorticity at boundaries requires special treatment (e.g., Thom's
formula)
% For contour plotting, usually omitting boundaries or simple extrapolation is
okay.

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% Plot final vorticity for this Re
figure(300 + re_idx); % Unique figure number for each Re vorticity plot
clf;
contourf(X, Y, omega, 50, 'LineColor', 'none'); % omega(i,j) corresponds to
Y(i), X(j)
colormap(jet);
colorbar;
title(sprintf('Vorticity (\\omega), Re = %d (Final State)', Re));
xlabel('x/L');
ylabel('y/L');
axis equal; axis([0 L 0 L]);
drawnow;

fprintf('--- Finished Simulation for Re = %d ---\\n', Re);

% Pause briefly to allow viewing plots before the next Re starts (optional)
% pause(1);

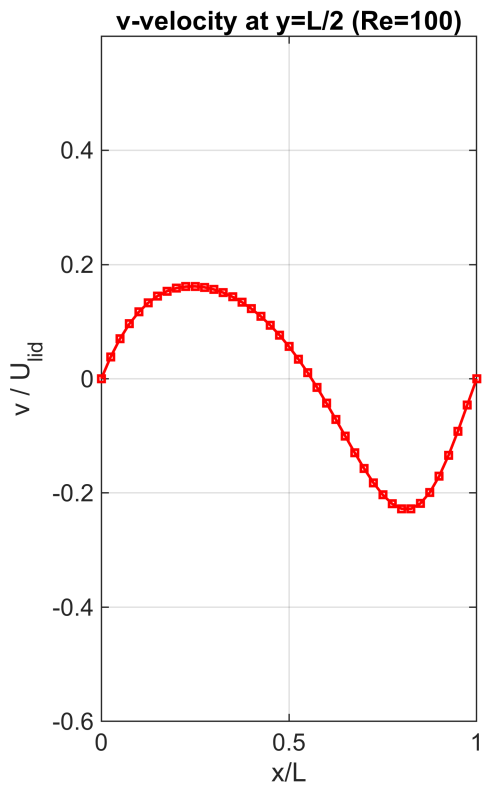
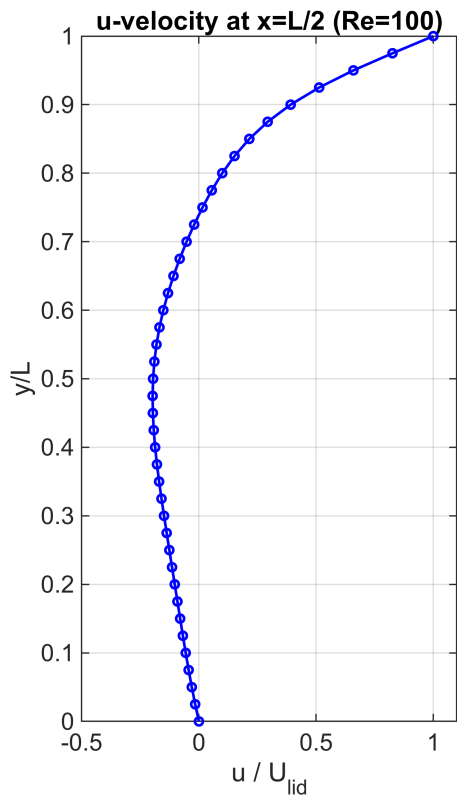
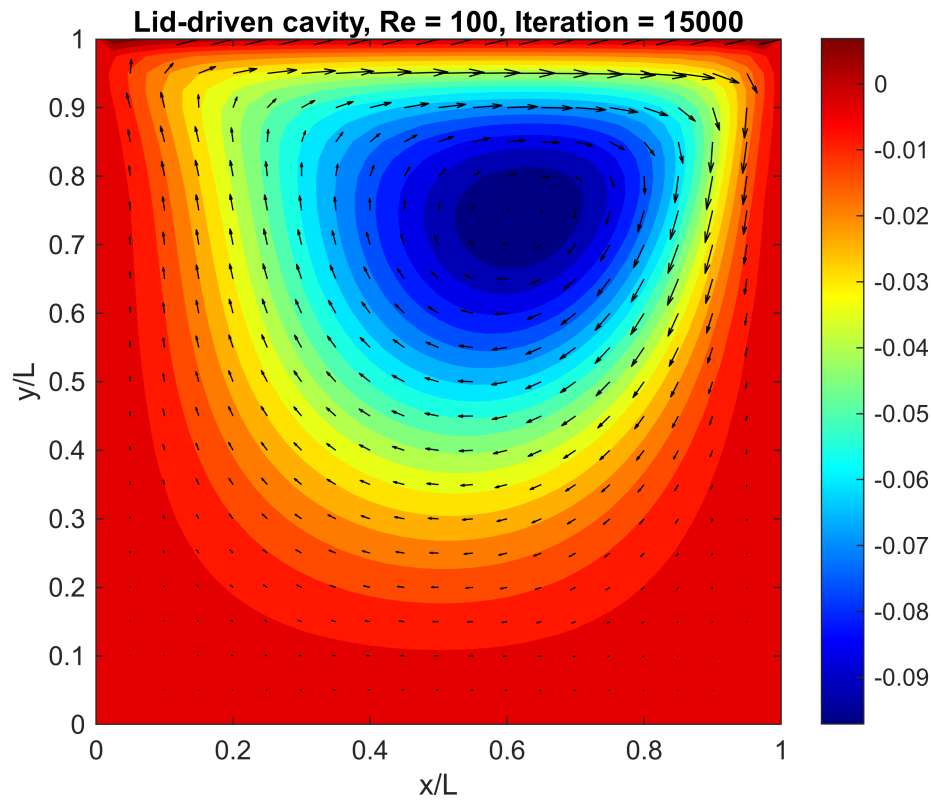
end % End of outer loop (Re_values)

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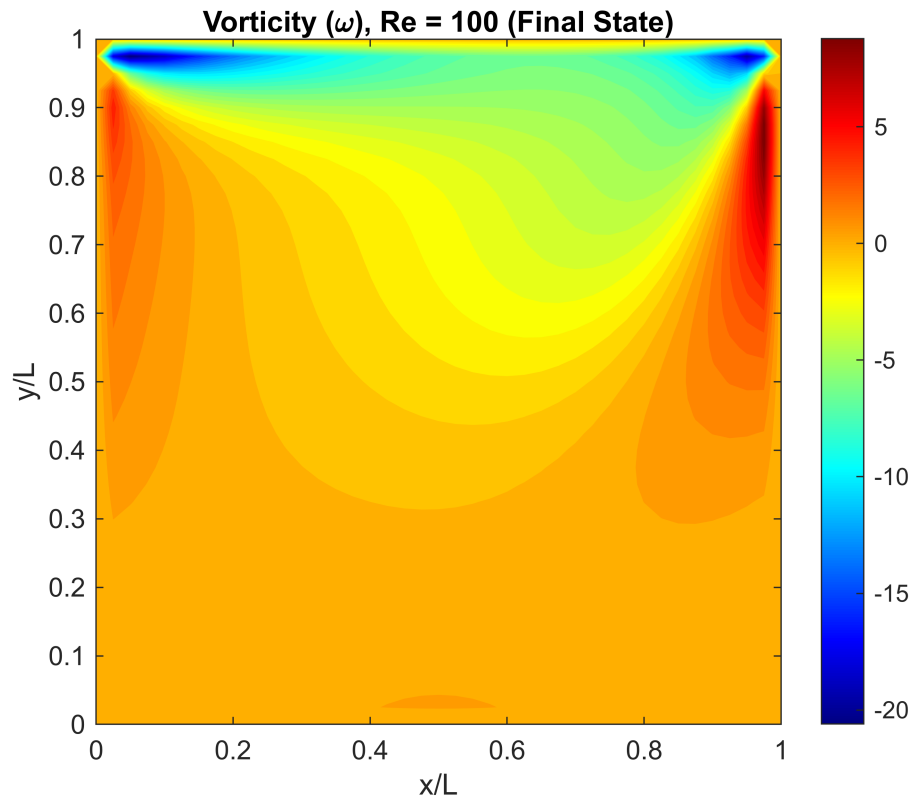
--- Starting Simulation for Re = 100 ---
Re=100, Iteration 500/15000, Max Vel Change = 4.172724e-04
Re=100, Iteration 1000/15000, Max Vel Change = 2.066710e-04
Re=100, Iteration 1500/15000, Max Vel Change = 1.287652e-04
Re=100, Iteration 2000/15000, Max Vel Change = 8.950378e-05
Re=100, Iteration 2500/15000, Max Vel Change = 6.529600e-05
Re=100, Iteration 3000/15000, Max Vel Change = 4.936963e-05
Re=100, Iteration 3500/15000, Max Vel Change = 3.868500e-05
Re=100, Iteration 4000/15000, Max Vel Change = 3.120416e-05
Re=100, Iteration 4500/15000, Max Vel Change = 2.552402e-05
Re=100, Iteration 5000/15000, Max Vel Change = 2.120261e-05
Re=100, Iteration 5500/15000, Max Vel Change = 1.786336e-05
Re=100, Iteration 6000/15000, Max Vel Change = 1.512103e-05
Re=100, Iteration 6500/15000, Max Vel Change = 1.287037e-05
Re=100, Iteration 7000/15000, Max Vel Change = 1.102139e-05
Re=100, Iteration 7500/15000, Max Vel Change = 9.465727e-06
Re=100, Iteration 8000/15000, Max Vel Change = 8.172011e-06
Re=100, Iteration 8500/15000, Max Vel Change = 7.068277e-06
Re=100, Iteration 9000/15000, Max Vel Change = 6.127397e-06
Re=100, Iteration 9500/15000, Max Vel Change = 5.326365e-06
Re=100, Iteration 10000/15000, Max Vel Change = 4.633769e-06
Re=100, Iteration 10500/15000, Max Vel Change = 4.034240e-06
Re=100, Iteration 11000/15000, Max Vel Change = 3.515221e-06
Re=100, Iteration 11500/15000, Max Vel Change = 3.068137e-06
Re=100, Iteration 12000/15000, Max Vel Change = 2.678790e-06
Re=100, Iteration 12500/15000, Max Vel Change = 2.339523e-06
Re=100, Iteration 13000/15000, Max Vel Change = 2.043733e-06
Re=100, Iteration 13500/15000, Max Vel Change = 1.785719e-06
Re=100, Iteration 14000/15000, Max Vel Change = 1.560558e-06
Re=100, Iteration 14500/15000, Max Vel Change = 1.363991e-06
Re=100, Iteration 15000/15000, Max Vel Change = 1.192331e-06

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Simulation time for $Re=100$: 289.63 seconds

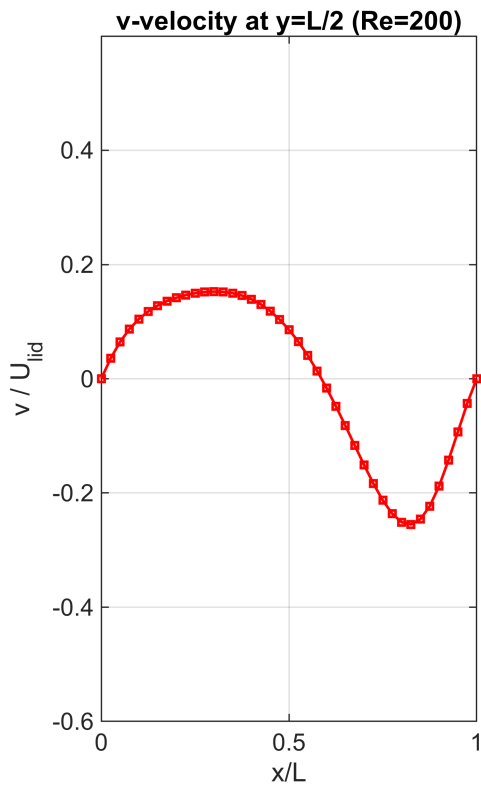
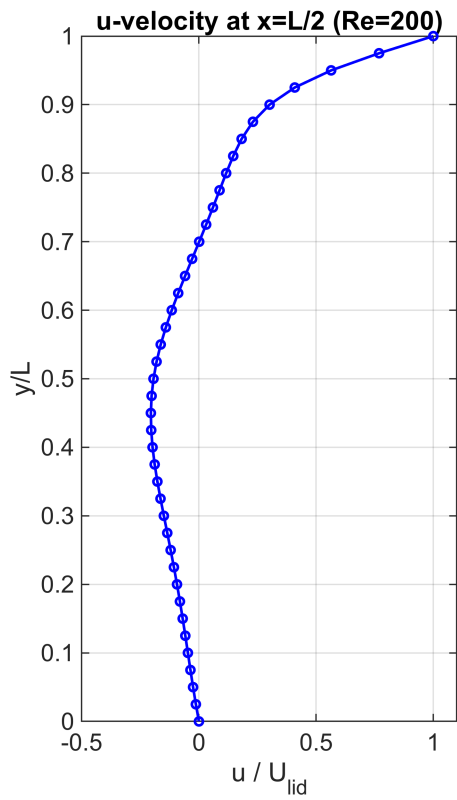
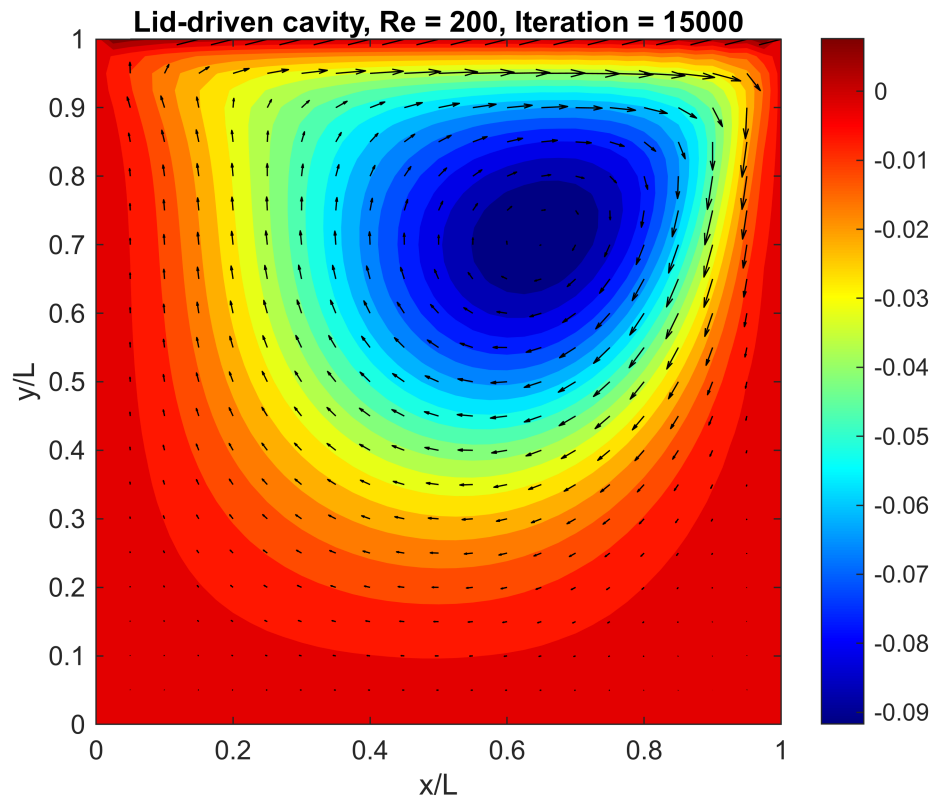
Warning: Solution for $Re=100$ did not fully converge within 15000 iterations. Final Max Vel Change = $1.192331e-06$



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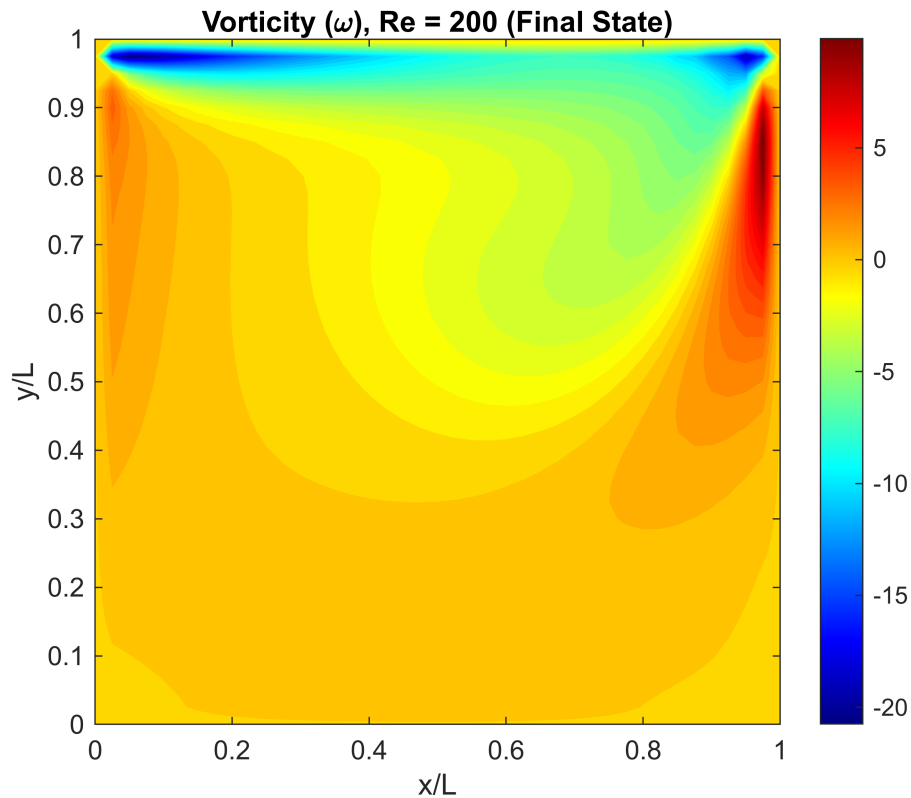
--- Finished Simulation for Re = 100 ---
--- Starting Simulation for Re = 200 ---
Re=200, Iteration 500/15000, Max Vel Change = 4.434089e-04
Re=200, Iteration 1000/15000, Max Vel Change = 2.302951e-04
Re=200, Iteration 1500/15000, Max Vel Change = 1.518947e-04
Re=200, Iteration 2000/15000, Max Vel Change = 1.148629e-04
Re=200, Iteration 2500/15000, Max Vel Change = 9.264102e-05
Re=200, Iteration 3000/15000, Max Vel Change = 7.625911e-05
Re=200, Iteration 3500/15000, Max Vel Change = 6.264569e-05
Re=200, Iteration 4000/15000, Max Vel Change = 5.280011e-05
Re=200, Iteration 4500/15000, Max Vel Change = 4.475598e-05
Re=200, Iteration 5000/15000, Max Vel Change = 3.855019e-05
Re=200, Iteration 5500/15000, Max Vel Change = 3.325839e-05
Re=200, Iteration 6000/15000, Max Vel Change = 2.915875e-05
Re=200, Iteration 6500/15000, Max Vel Change = 2.576848e-05
Re=200, Iteration 7000/15000, Max Vel Change = 2.287996e-05
Re=200, Iteration 7500/15000, Max Vel Change = 2.050676e-05
Re=200, Iteration 8000/15000, Max Vel Change = 1.842629e-05
Re=200, Iteration 8500/15000, Max Vel Change = 1.688955e-05
Re=200, Iteration 9000/15000, Max Vel Change = 1.552979e-05
Re=200, Iteration 9500/15000, Max Vel Change = 1.433818e-05
Re=200, Iteration 10000/15000, Max Vel Change = 1.314644e-05
Re=200, Iteration 10500/15000, Max Vel Change = 1.216710e-05
Re=200, Iteration 11000/15000, Max Vel Change = 1.125151e-05
Re=200, Iteration 11500/15000, Max Vel Change = 1.036358e-05
Re=200, Iteration 12000/15000, Max Vel Change = 9.547635e-06
Re=200, Iteration 12500/15000, Max Vel Change = 8.862464e-06
Re=200, Iteration 13000/15000, Max Vel Change = 8.206936e-06
Re=200, Iteration 13500/15000, Max Vel Change = 7.586547e-06
Re=200, Iteration 14000/15000, Max Vel Change = 7.004268e-06
Re=200, Iteration 14500/15000, Max Vel Change = 6.486089e-06
Re=200, Iteration 15000/15000, Max Vel Change = 6.023758e-06

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Simulation time for Re=200: 289.04 seconds

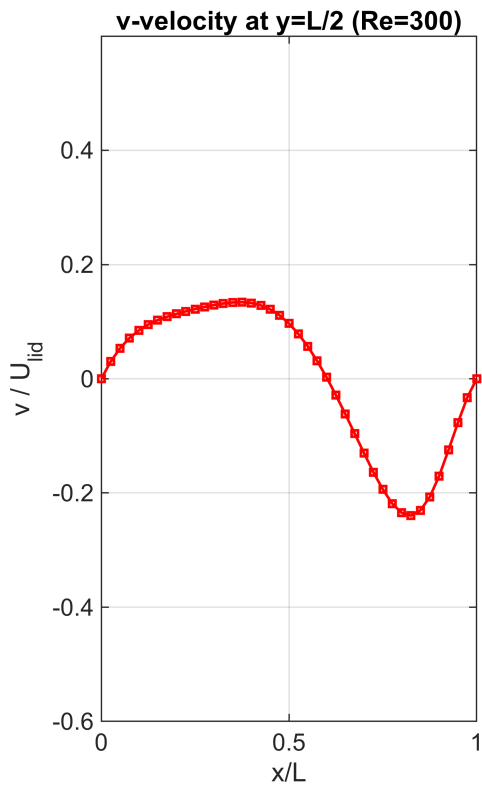
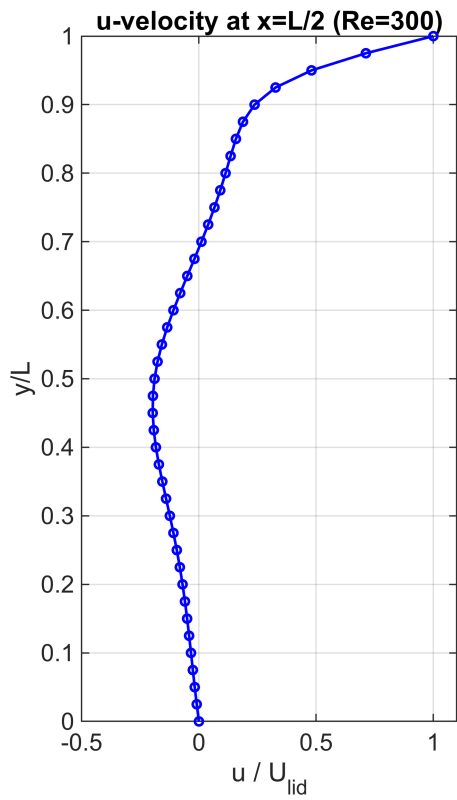
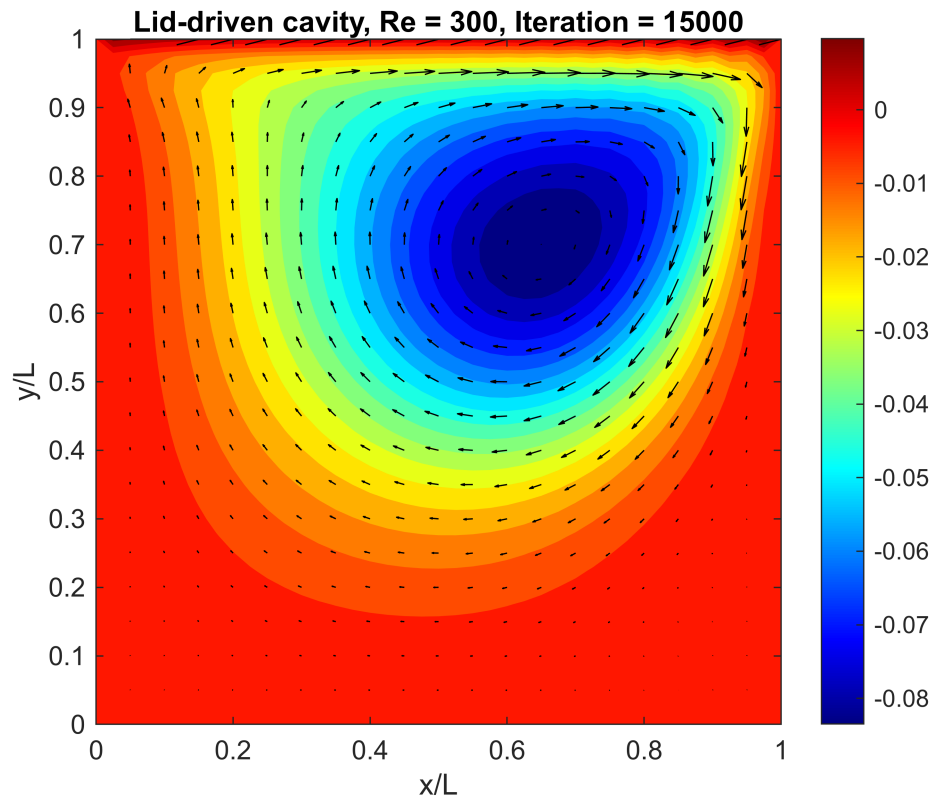
Warning: Solution for Re=200 did not fully converge within 15000 iterations. Final Max Vel Change = $6.023758e-06$



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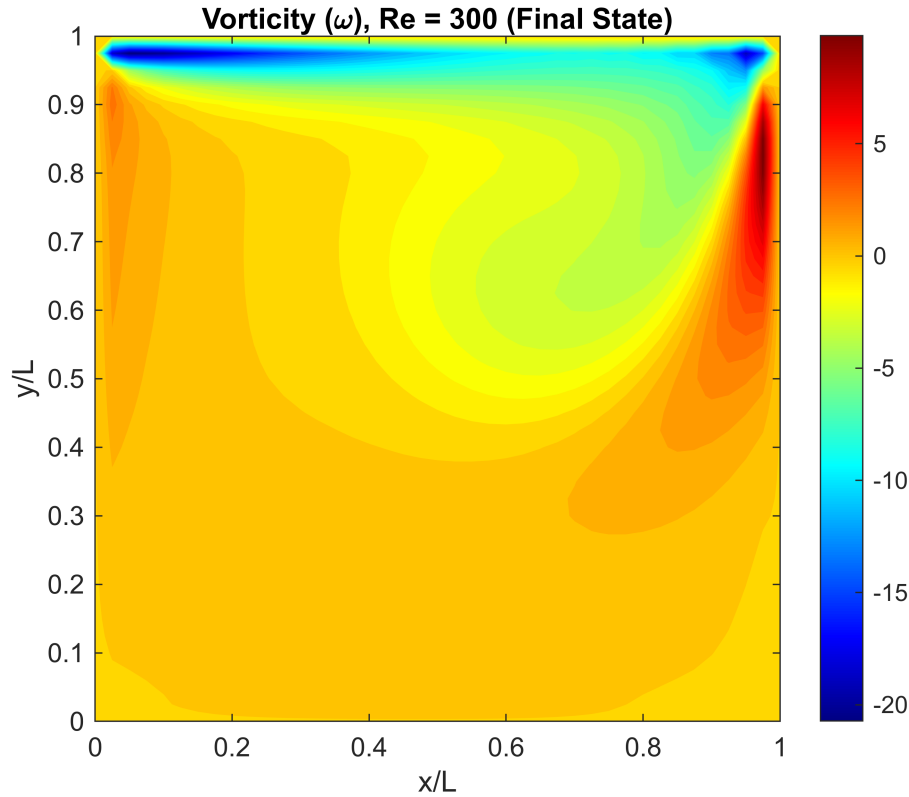
--- Finished Simulation for Re = 200 ---
--- Starting Simulation for Re = 300 ---
Re=300, Iteration 500/15000, Max Vel Change = 4.195403e-04
Re=300, Iteration 1000/15000, Max Vel Change = 2.383702e-04
Re=300, Iteration 1500/15000, Max Vel Change = 1.640766e-04
Re=300, Iteration 2000/15000, Max Vel Change = 1.265261e-04
Re=300, Iteration 2500/15000, Max Vel Change = 1.067727e-04
Re=300, Iteration 3000/15000, Max Vel Change = 8.910806e-05
Re=300, Iteration 3500/15000, Max Vel Change = 7.429189e-05
Re=300, Iteration 4000/15000, Max Vel Change = 6.306825e-05
Re=300, Iteration 4500/15000, Max Vel Change = 5.441963e-05
Re=300, Iteration 5000/15000, Max Vel Change = 4.678959e-05
Re=300, Iteration 5500/15000, Max Vel Change = 4.138411e-05
Re=300, Iteration 6000/15000, Max Vel Change = 3.638333e-05
Re=300, Iteration 6500/15000, Max Vel Change = 3.294854e-05
Re=300, Iteration 7000/15000, Max Vel Change = 3.044114e-05
Re=300, Iteration 7500/15000, Max Vel Change = 2.817633e-05
Re=300, Iteration 8000/15000, Max Vel Change = 2.637901e-05
Re=300, Iteration 8500/15000, Max Vel Change = 2.439776e-05
Re=300, Iteration 9000/15000, Max Vel Change = 2.297910e-05
Re=300, Iteration 9500/15000, Max Vel Change = 2.136471e-05
Re=300, Iteration 10000/15000, Max Vel Change = 2.007500e-05
Re=300, Iteration 10500/15000, Max Vel Change = 1.882423e-05
Re=300, Iteration 11000/15000, Max Vel Change = 1.752180e-05
Re=300, Iteration 11500/15000, Max Vel Change = 1.654993e-05
Re=300, Iteration 12000/15000, Max Vel Change = 1.550320e-05
Re=300, Iteration 12500/15000, Max Vel Change = 1.446304e-05
Re=300, Iteration 13000/15000, Max Vel Change = 1.364661e-05
Re=300, Iteration 13500/15000, Max Vel Change = 1.282938e-05
Re=300, Iteration 14000/15000, Max Vel Change = 1.199903e-05
Re=300, Iteration 14500/15000, Max Vel Change = 1.125983e-05
Re=300, Iteration 15000/15000, Max Vel Change = 1.060831e-05

```



Simulation time for $Re=300$: 272.52 seconds

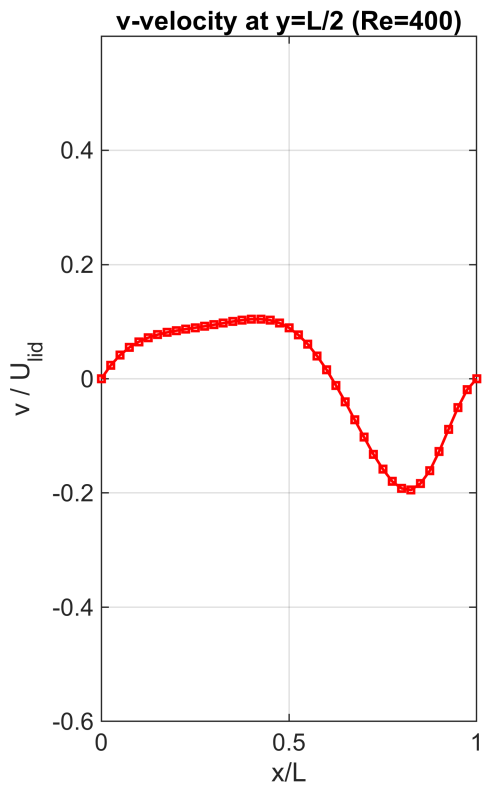
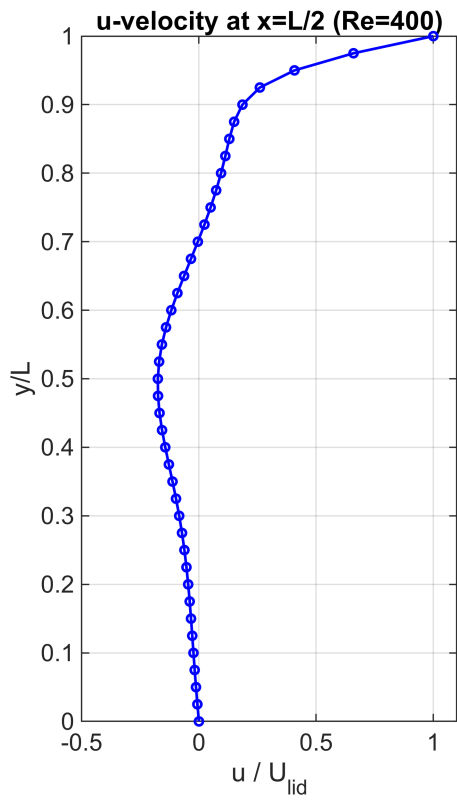
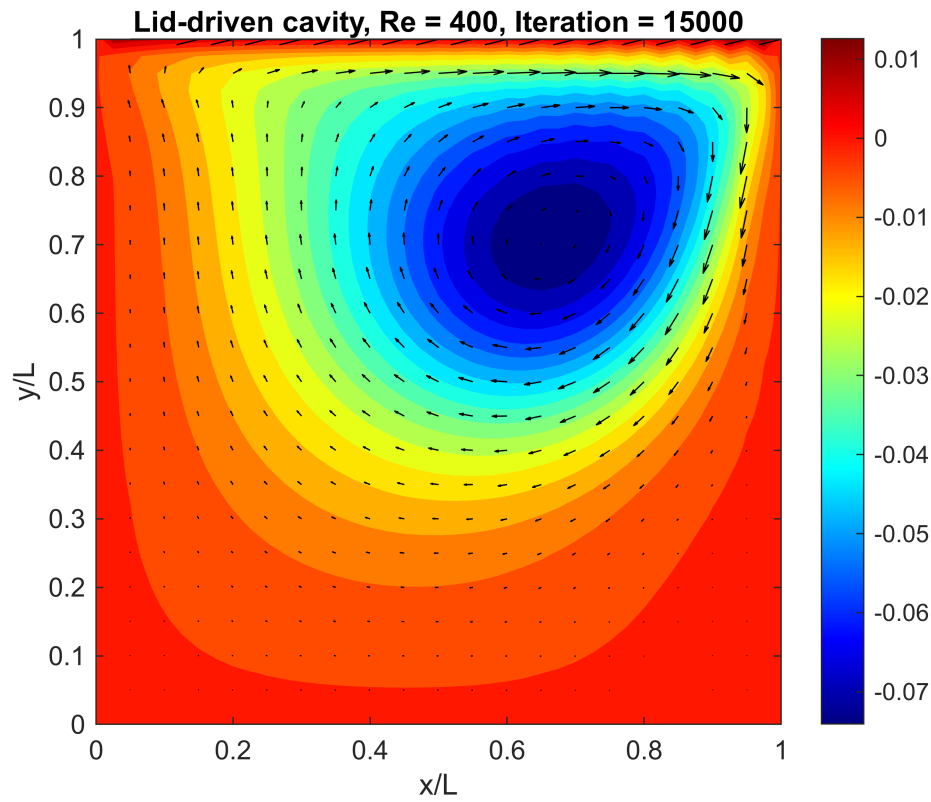
Warning: Solution for $Re=300$ did not fully converge within 15000 iterations. Final Max Vel Change = $1.060831e-05$



```

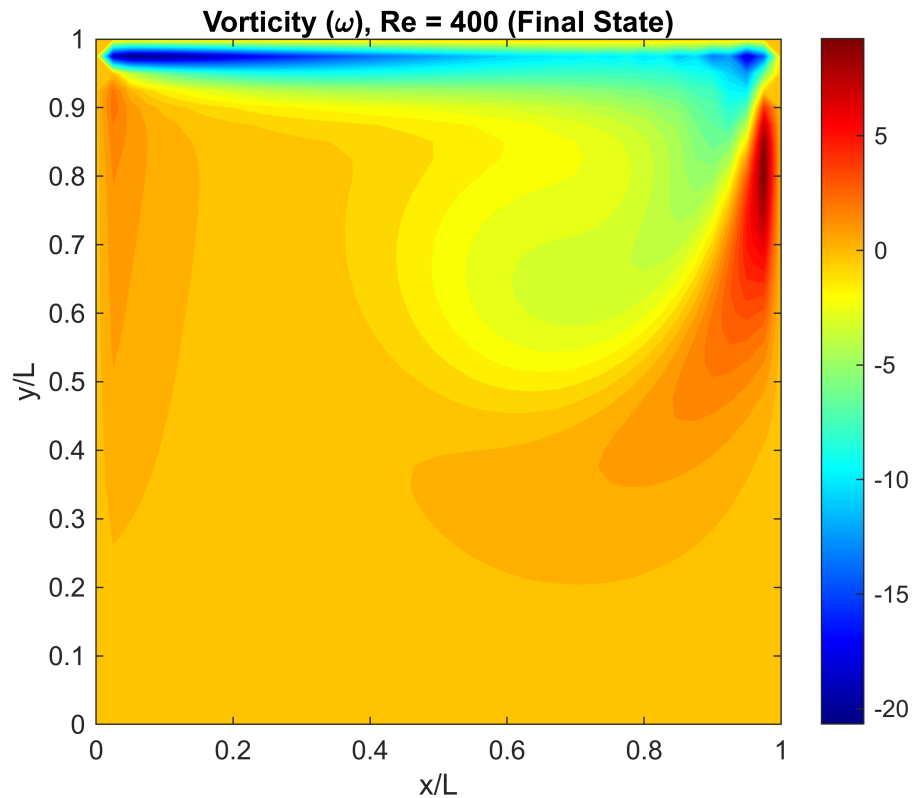
--- Finished Simulation for Re = 300 ---
--- Starting Simulation for Re = 400 ---
Re=400, Iteration 500/15000, Max Vel Change = 4.826893e-04
Re=400, Iteration 1000/15000, Max Vel Change = 2.570413e-04
Re=400, Iteration 1500/15000, Max Vel Change = 1.701671e-04
Re=400, Iteration 2000/15000, Max Vel Change = 1.274767e-04
Re=400, Iteration 2500/15000, Max Vel Change = 1.085476e-04
Re=400, Iteration 3000/15000, Max Vel Change = 9.246621e-05
Re=400, Iteration 3500/15000, Max Vel Change = 7.825255e-05
Re=400, Iteration 4000/15000, Max Vel Change = 6.593343e-05
Re=400, Iteration 4500/15000, Max Vel Change = 5.684080e-05
Re=400, Iteration 5000/15000, Max Vel Change = 4.965500e-05
Re=400, Iteration 5500/15000, Max Vel Change = 4.340884e-05
Re=400, Iteration 6000/15000, Max Vel Change = 3.982357e-05
Re=400, Iteration 6500/15000, Max Vel Change = 3.699618e-05
Re=400, Iteration 7000/15000, Max Vel Change = 3.469332e-05
Re=400, Iteration 7500/15000, Max Vel Change = 3.182044e-05
Re=400, Iteration 8000/15000, Max Vel Change = 3.030160e-05
Re=400, Iteration 8500/15000, Max Vel Change = 2.811310e-05
Re=400, Iteration 9000/15000, Max Vel Change = 2.667467e-05
Re=400, Iteration 9500/15000, Max Vel Change = 2.485746e-05
Re=400, Iteration 10000/15000, Max Vel Change = 2.360876e-05
Re=400, Iteration 10500/15000, Max Vel Change = 2.213091e-05
Re=400, Iteration 11000/15000, Max Vel Change = 2.098985e-05
Re=400, Iteration 11500/15000, Max Vel Change = 1.983616e-05
Re=400, Iteration 12000/15000, Max Vel Change = 1.860073e-05
Re=400, Iteration 12500/15000, Max Vel Change = 1.772829e-05
Re=400, Iteration 13000/15000, Max Vel Change = 1.675830e-05
Re=400, Iteration 13500/15000, Max Vel Change = 1.576092e-05
Re=400, Iteration 14000/15000, Max Vel Change = 1.500841e-05
Re=400, Iteration 14500/15000, Max Vel Change = 1.423349e-05
Re=400, Iteration 15000/15000, Max Vel Change = 1.345351e-05

```



Simulation time for Re=400: 266.74 seconds

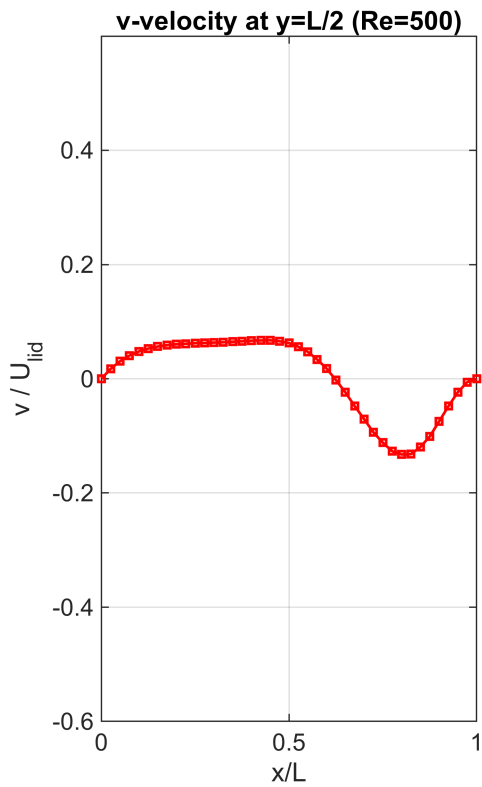
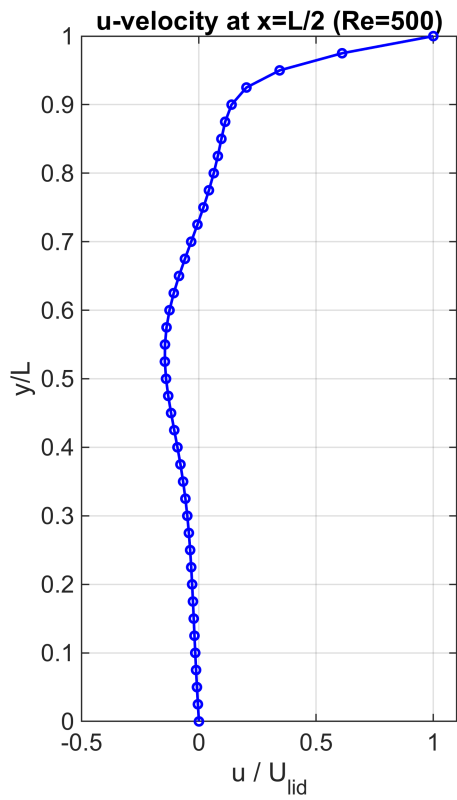
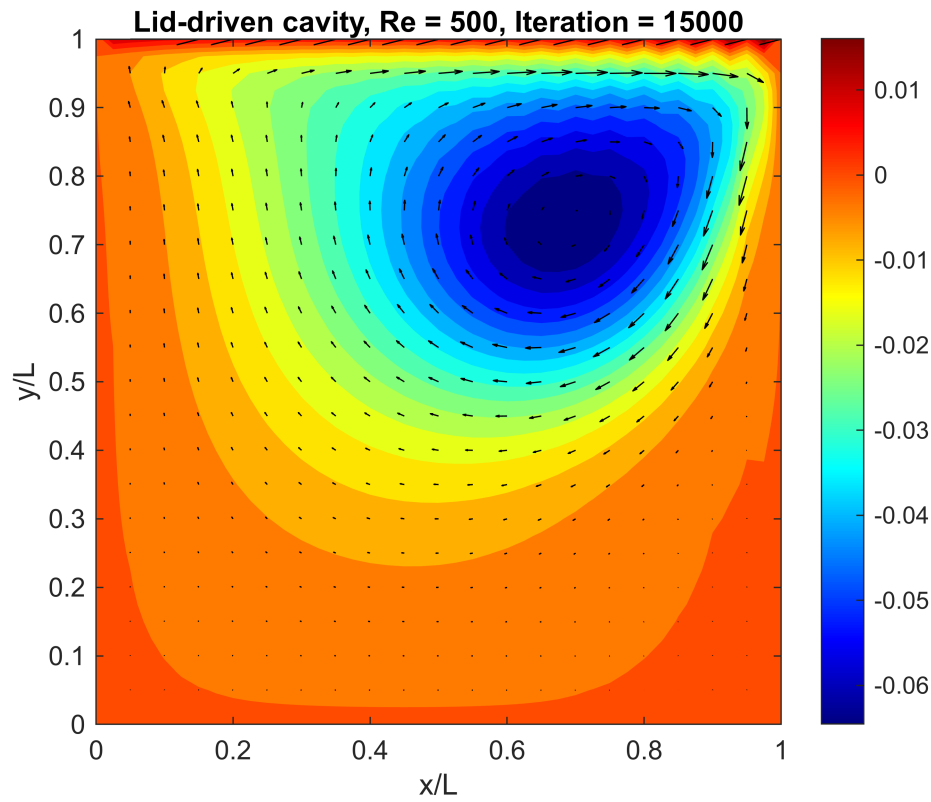
Warning: Solution for Re=400 did not fully converge within 15000 iterations. Final Max Vel Change = 1.345351e-05



```

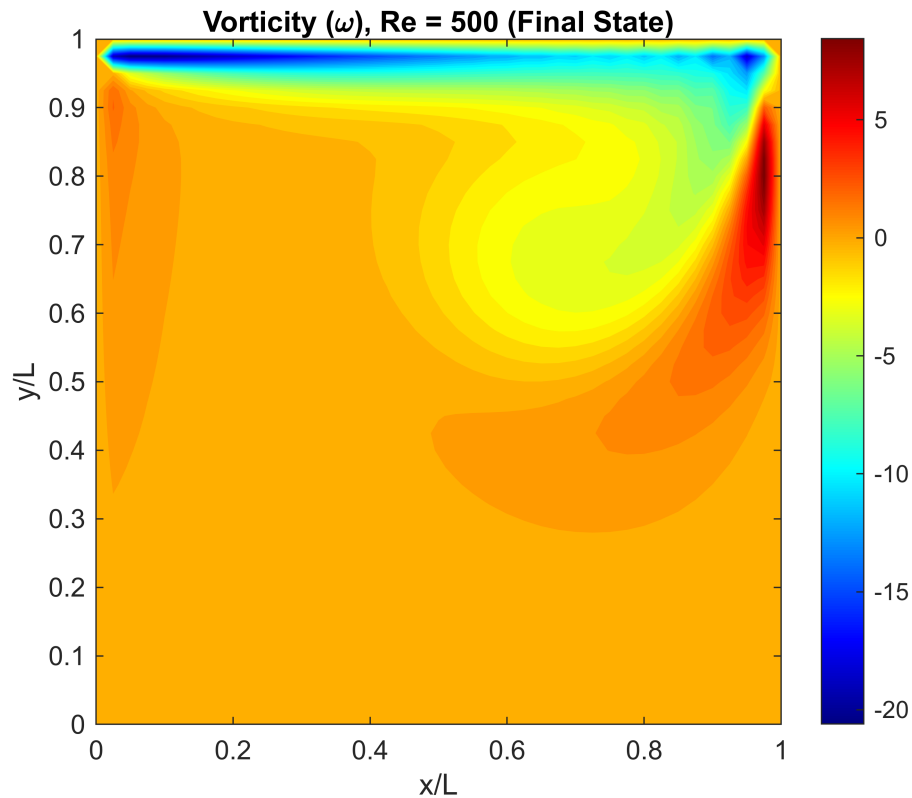
--- Finished Simulation for Re = 400 ---
--- Starting Simulation for Re = 500 ---
Re=500, Iteration 500/15000, Max Vel Change = 5.352227e-04
Re=500, Iteration 1000/15000, Max Vel Change = 2.890243e-04
Re=500, Iteration 1500/15000, Max Vel Change = 1.826049e-04
Re=500, Iteration 2000/15000, Max Vel Change = 1.300032e-04
Re=500, Iteration 2500/15000, Max Vel Change = 1.057846e-04
Re=500, Iteration 3000/15000, Max Vel Change = 8.949676e-05
Re=500, Iteration 3500/15000, Max Vel Change = 7.654247e-05
Re=500, Iteration 4000/15000, Max Vel Change = 6.480184e-05
Re=500, Iteration 4500/15000, Max Vel Change = 5.498901e-05
Re=500, Iteration 5000/15000, Max Vel Change = 4.798408e-05
Re=500, Iteration 5500/15000, Max Vel Change = 4.376310e-05
Re=500, Iteration 6000/15000, Max Vel Change = 4.086926e-05
Re=500, Iteration 6500/15000, Max Vel Change = 3.747044e-05
Re=500, Iteration 7000/15000, Max Vel Change = 3.518071e-05
Re=500, Iteration 7500/15000, Max Vel Change = 3.275178e-05
Re=500, Iteration 8000/15000, Max Vel Change = 3.071492e-05
Re=500, Iteration 8500/15000, Max Vel Change = 2.875644e-05
Re=500, Iteration 9000/15000, Max Vel Change = 2.720265e-05
Re=500, Iteration 9500/15000, Max Vel Change = 2.548536e-05
Re=500, Iteration 10000/15000, Max Vel Change = 2.422796e-05
Re=500, Iteration 10500/15000, Max Vel Change = 2.265489e-05
Re=500, Iteration 11000/15000, Max Vel Change = 2.162815e-05
Re=500, Iteration 11500/15000, Max Vel Change = 2.051402e-05
Re=500, Iteration 12000/15000, Max Vel Change = 1.925196e-05
Re=500, Iteration 12500/15000, Max Vel Change = 1.844754e-05
Re=500, Iteration 13000/15000, Max Vel Change = 1.752200e-05
Re=500, Iteration 13500/15000, Max Vel Change = 1.655118e-05
Re=500, Iteration 14000/15000, Max Vel Change = 1.579644e-05
Re=500, Iteration 14500/15000, Max Vel Change = 1.507514e-05
Re=500, Iteration 15000/15000, Max Vel Change = 1.433012e-05

```



Simulation time for Re=500: 268.89 seconds

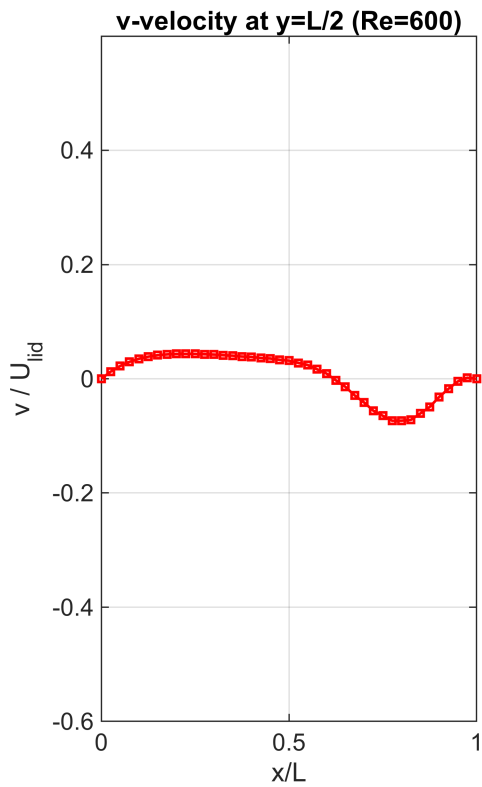
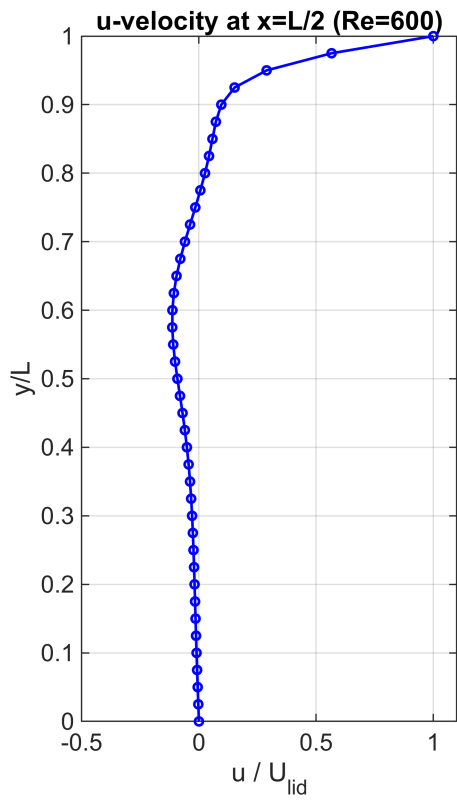
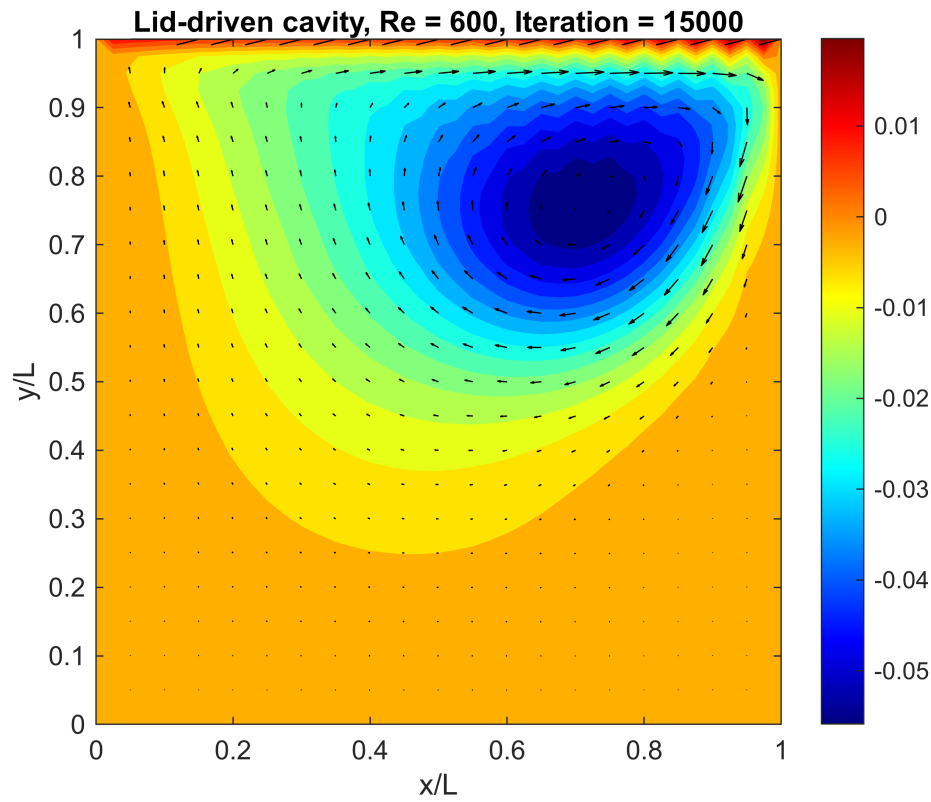
Warning: Solution for Re=500 did not fully converge within 15000 iterations. Final Max Vel Change = $1.433012e-05$



```

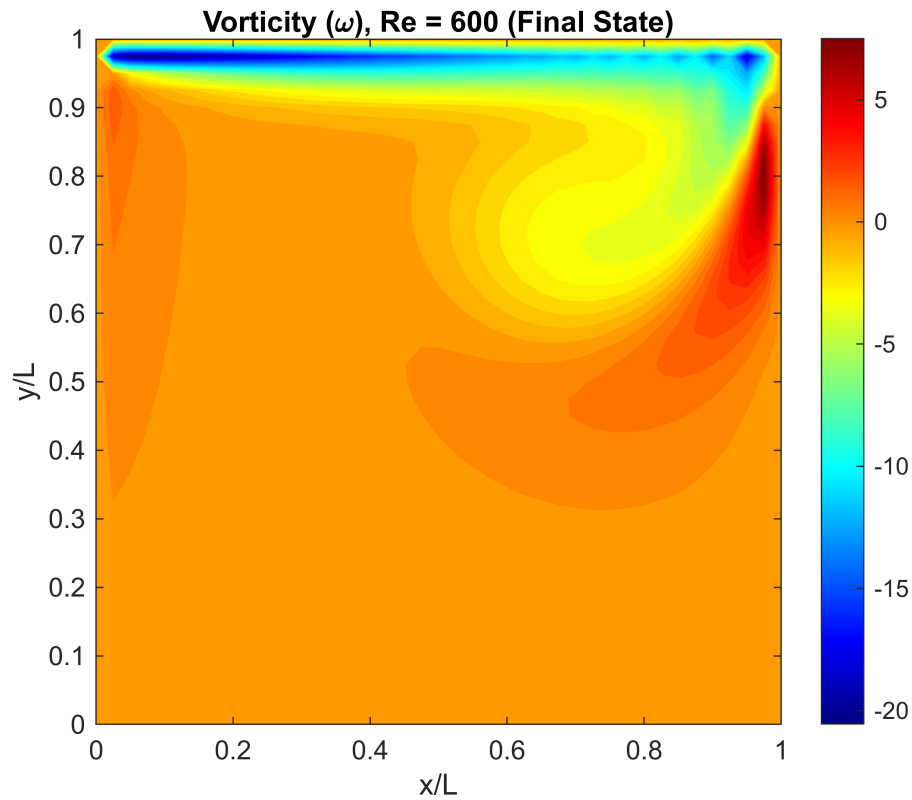
--- Finished Simulation for Re = 500 ---
--- Starting Simulation for Re = 600 ---
Re=600, Iteration 500/15000, Max Vel Change = 5.653120e-04
Re=600, Iteration 1000/15000, Max Vel Change = 3.138938e-04
Re=600, Iteration 1500/15000, Max Vel Change = 1.980425e-04
Re=600, Iteration 2000/15000, Max Vel Change = 1.435322e-04
Re=600, Iteration 2500/15000, Max Vel Change = 1.062569e-04
Re=600, Iteration 3000/15000, Max Vel Change = 8.731983e-05
Re=600, Iteration 3500/15000, Max Vel Change = 7.158053e-05
Re=600, Iteration 4000/15000, Max Vel Change = 6.119671e-05
Re=600, Iteration 4500/15000, Max Vel Change = 5.210439e-05
Re=600, Iteration 5000/15000, Max Vel Change = 4.537641e-05
Re=600, Iteration 5500/15000, Max Vel Change = 4.264126e-05
Re=600, Iteration 6000/15000, Max Vel Change = 3.840659e-05
Re=600, Iteration 6500/15000, Max Vel Change = 3.606304e-05
Re=600, Iteration 7000/15000, Max Vel Change = 3.306419e-05
Re=600, Iteration 7500/15000, Max Vel Change = 3.125855e-05
Re=600, Iteration 8000/15000, Max Vel Change = 2.877501e-05
Re=600, Iteration 8500/15000, Max Vel Change = 2.739866e-05
Re=600, Iteration 9000/15000, Max Vel Change = 2.538724e-05
Re=600, Iteration 9500/15000, Max Vel Change = 2.409223e-05
Re=600, Iteration 10000/15000, Max Vel Change = 2.286796e-05
Re=600, Iteration 10500/15000, Max Vel Change = 2.125746e-05
Re=600, Iteration 11000/15000, Max Vel Change = 2.033554e-05
Re=600, Iteration 11500/15000, Max Vel Change = 1.935591e-05
Re=600, Iteration 12000/15000, Max Vel Change = 1.815662e-05
Re=600, Iteration 12500/15000, Max Vel Change = 1.728872e-05
Re=600, Iteration 13000/15000, Max Vel Change = 1.652888e-05
Re=600, Iteration 13500/15000, Max Vel Change = 1.569806e-05
Re=600, Iteration 14000/15000, Max Vel Change = 1.479250e-05
Re=600, Iteration 14500/15000, Max Vel Change = 1.413218e-05
Re=600, Iteration 15000/15000, Max Vel Change = 1.351927e-05

```



Simulation time for Re=600: 290.30 seconds

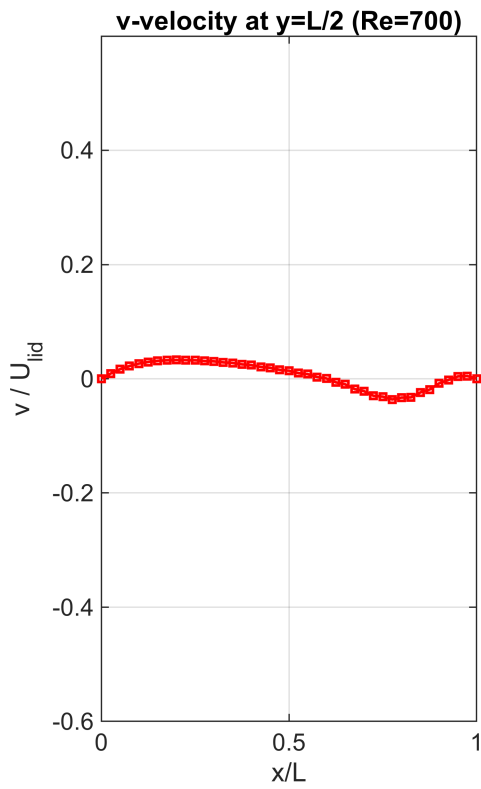
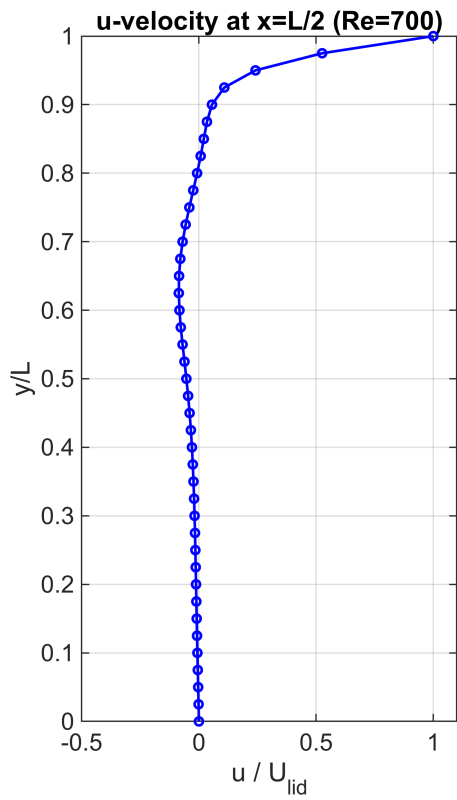
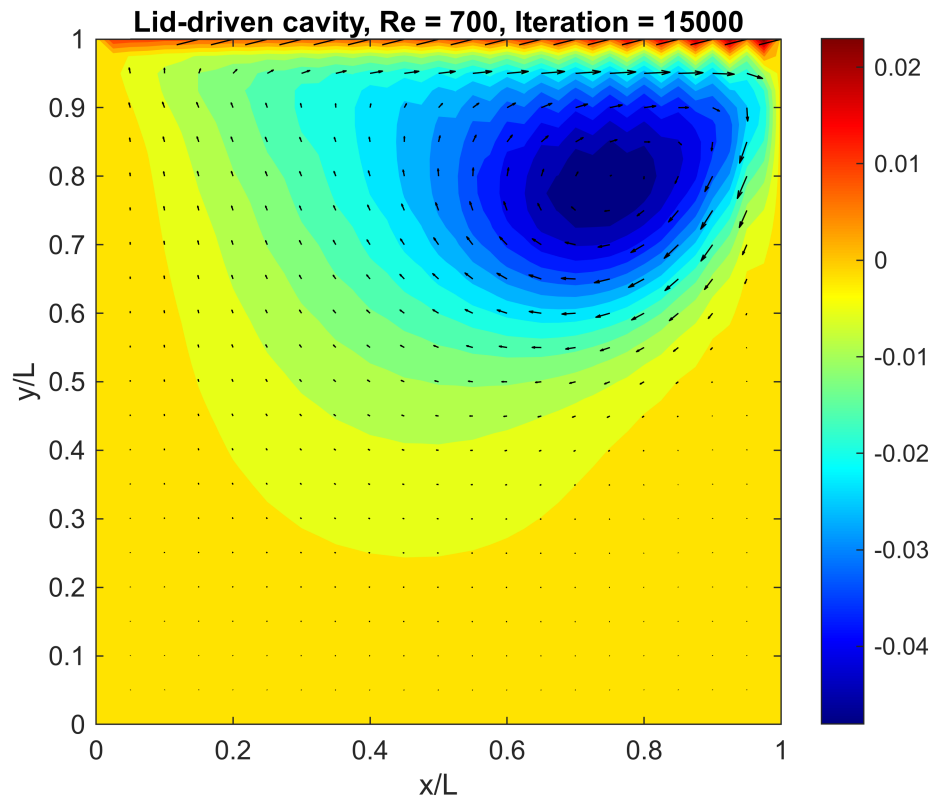
Warning: Solution for Re=600 did not fully converge within 15000 iterations. Final Max Vel Change = 1.351927e-05



```

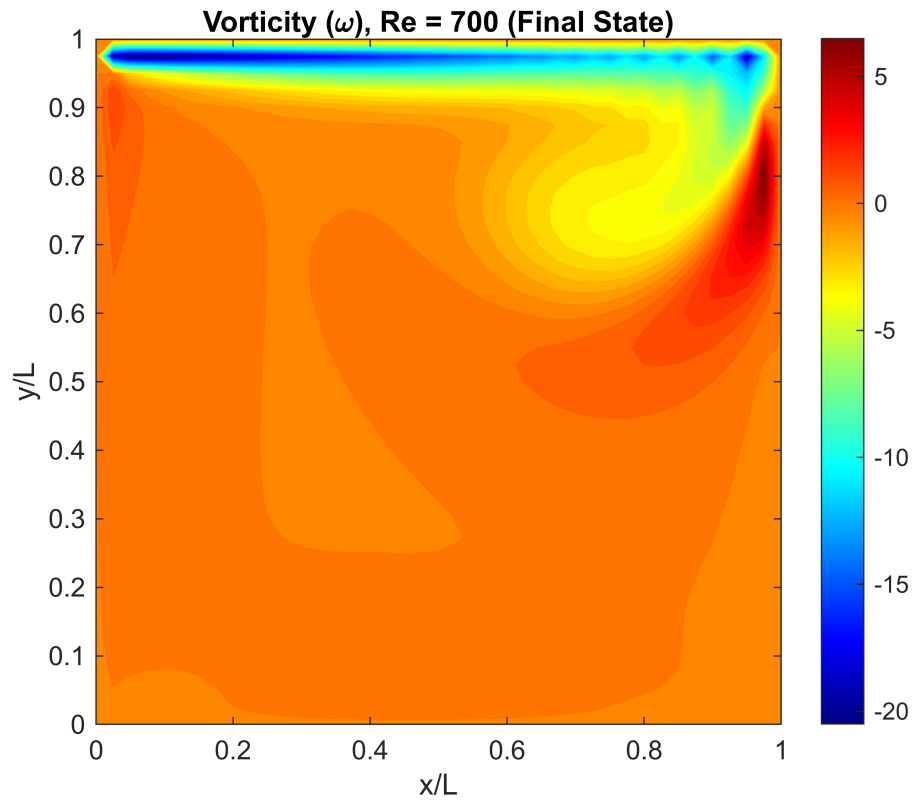
--- Finished Simulation for Re = 600 ---
--- Starting Simulation for Re = 700 ---
Re=700, Iteration 500/15000, Max Vel Change = 5.893802e-04
Re=700, Iteration 1000/15000, Max Vel Change = 3.410590e-04
Re=700, Iteration 1500/15000, Max Vel Change = 2.164487e-04
Re=700, Iteration 2000/15000, Max Vel Change = 1.548388e-04
Re=700, Iteration 2500/15000, Max Vel Change = 1.151308e-04
Re=700, Iteration 3000/15000, Max Vel Change = 8.714396e-05
Re=700, Iteration 3500/15000, Max Vel Change = 6.986616e-05
Re=700, Iteration 4000/15000, Max Vel Change = 5.670740e-05
Re=700, Iteration 4500/15000, Max Vel Change = 4.822843e-05
Re=700, Iteration 5000/15000, Max Vel Change = 4.212977e-05
Re=700, Iteration 5500/15000, Max Vel Change = 3.910271e-05
Re=700, Iteration 6000/15000, Max Vel Change = 3.530883e-05
Re=700, Iteration 6500/15000, Max Vel Change = 3.296898e-05
Re=700, Iteration 7000/15000, Max Vel Change = 3.018809e-05
Re=700, Iteration 7500/15000, Max Vel Change = 2.817615e-05
Re=700, Iteration 8000/15000, Max Vel Change = 2.648113e-05
Re=700, Iteration 8500/15000, Max Vel Change = 2.430743e-05
Re=700, Iteration 9000/15000, Max Vel Change = 2.303199e-05
Re=700, Iteration 9500/15000, Max Vel Change = 2.176700e-05
Re=700, Iteration 10000/15000, Max Vel Change = 2.019509e-05
Re=700, Iteration 10500/15000, Max Vel Change = 1.910760e-05
Re=700, Iteration 11000/15000, Max Vel Change = 1.823815e-05
Re=700, Iteration 11500/15000, Max Vel Change = 1.718798e-05
Re=700, Iteration 12000/15000, Max Vel Change = 1.609042e-05
Re=700, Iteration 12500/15000, Max Vel Change = 1.521445e-05
Re=700, Iteration 13000/15000, Max Vel Change = 1.453249e-05
Re=700, Iteration 13500/15000, Max Vel Change = 1.376835e-05
Re=700, Iteration 14000/15000, Max Vel Change = 1.302173e-05
Re=700, Iteration 14500/15000, Max Vel Change = 1.230160e-05
Re=700, Iteration 15000/15000, Max Vel Change = 1.159838e-05

```



Simulation time for Re=700: 281.70 seconds

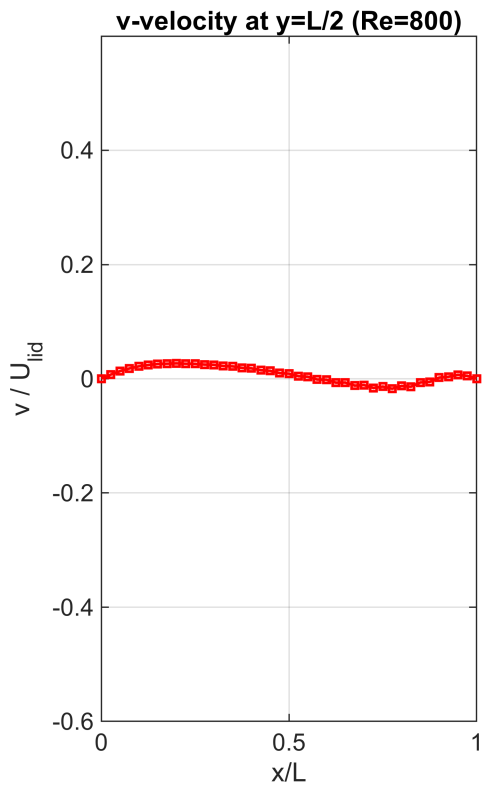
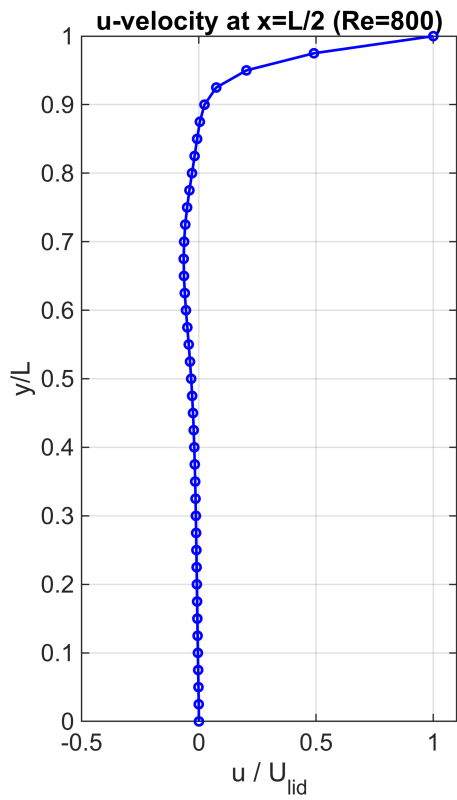
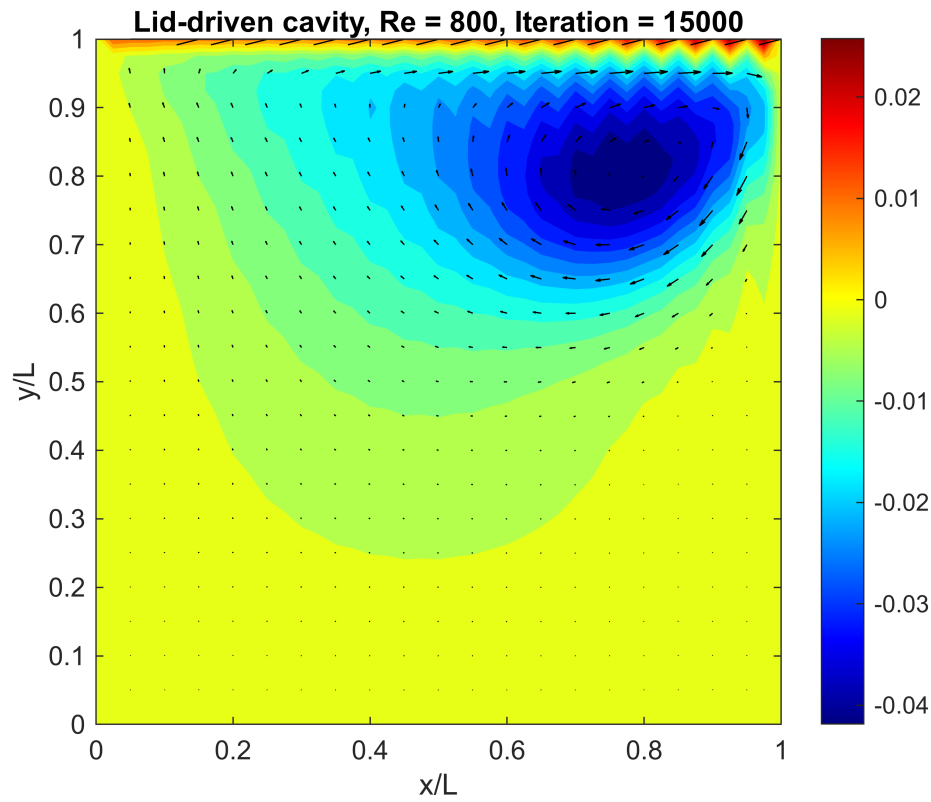
Warning: Solution for Re=700 did not fully converge within 15000 iterations. Final Max Vel Change = 1.159838e-05



```

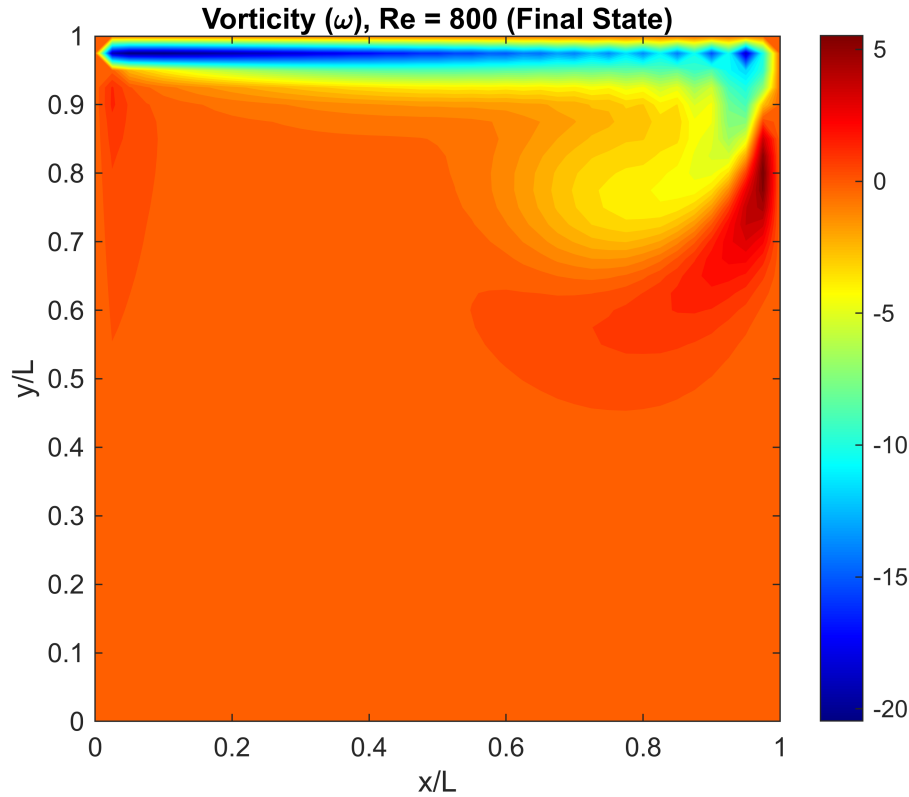
--- Finished Simulation for Re = 700 ---
--- Starting Simulation for Re = 800 ---
Re=800, Iteration 500/15000, Max Vel Change = 6.133327e-04
Re=800, Iteration 1000/15000, Max Vel Change = 3.760927e-04
Re=800, Iteration 1500/15000, Max Vel Change = 2.441054e-04
Re=800, Iteration 2000/15000, Max Vel Change = 1.638405e-04
Re=800, Iteration 2500/15000, Max Vel Change = 1.185868e-04
Re=800, Iteration 3000/15000, Max Vel Change = 8.964207e-05
Re=800, Iteration 3500/15000, Max Vel Change = 6.742159e-05
Re=800, Iteration 4000/15000, Max Vel Change = 5.523930e-05
Re=800, Iteration 4500/15000, Max Vel Change = 4.610059e-05
Re=800, Iteration 5000/15000, Max Vel Change = 3.891495e-05
Re=800, Iteration 5500/15000, Max Vel Change = 3.536387e-05
Re=800, Iteration 6000/15000, Max Vel Change = 3.151950e-05
Re=800, Iteration 6500/15000, Max Vel Change = 2.972996e-05
Re=800, Iteration 7000/15000, Max Vel Change = 2.709937e-05
Re=800, Iteration 7500/15000, Max Vel Change = 2.474234e-05
Re=800, Iteration 8000/15000, Max Vel Change = 2.335116e-05
Re=800, Iteration 8500/15000, Max Vel Change = 2.165629e-05
Re=800, Iteration 9000/15000, Max Vel Change = 2.007155e-05
Re=800, Iteration 9500/15000, Max Vel Change = 1.893941e-05
Re=800, Iteration 10000/15000, Max Vel Change = 1.780437e-05
Re=800, Iteration 10500/15000, Max Vel Change = 1.672437e-05
Re=800, Iteration 11000/15000, Max Vel Change = 1.557680e-05
Re=800, Iteration 11500/15000, Max Vel Change = 1.455629e-05
Re=800, Iteration 12000/15000, Max Vel Change = 1.378432e-05
Re=800, Iteration 12500/15000, Max Vel Change = 1.307445e-05
Re=800, Iteration 13000/15000, Max Vel Change = 1.230740e-05
Re=800, Iteration 13500/15000, Max Vel Change = 1.151622e-05
Re=800, Iteration 14000/15000, Max Vel Change = 1.083062e-05
Re=800, Iteration 14500/15000, Max Vel Change = 1.013805e-05
Re=800, Iteration 15000/15000, Max Vel Change = 9.483813e-06

```



Simulation time for $Re=800$: 273.02 seconds

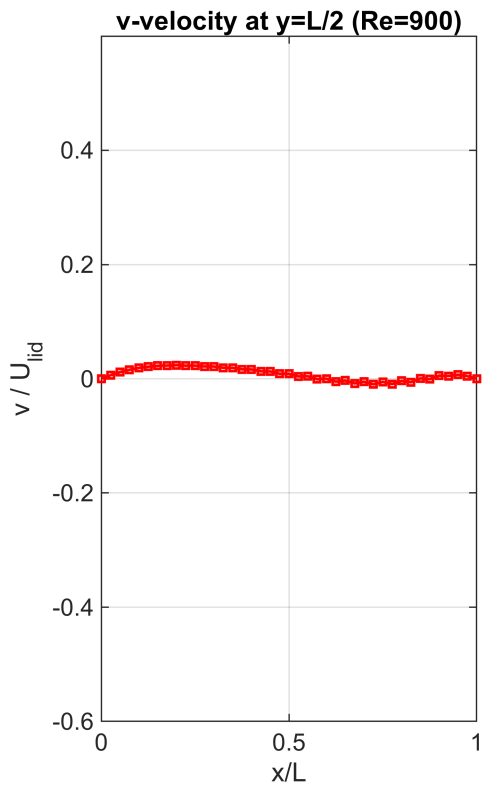
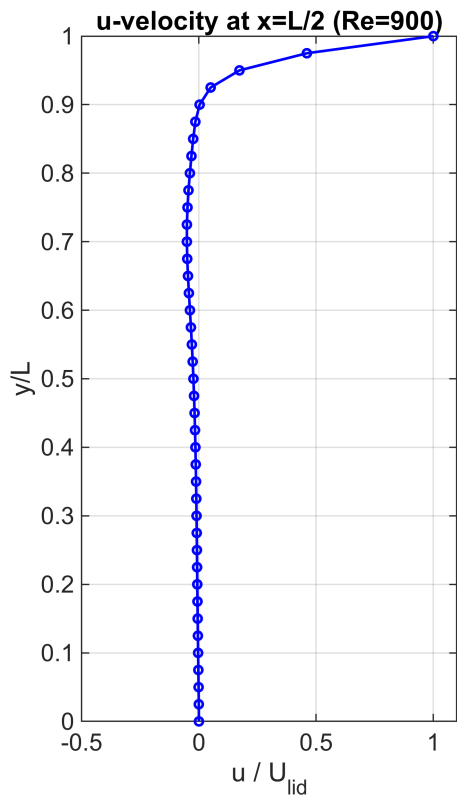
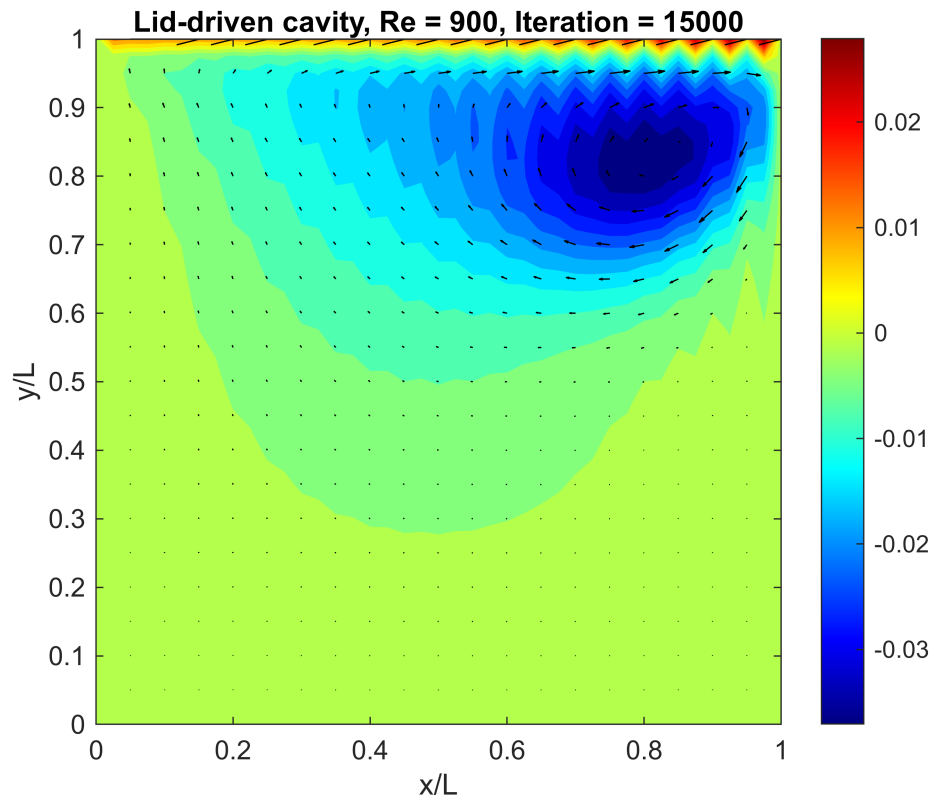
Warning: Solution for $Re=800$ did not fully converge within 15000 iterations. Final Max Vel Change = $9.483813e-06$



```

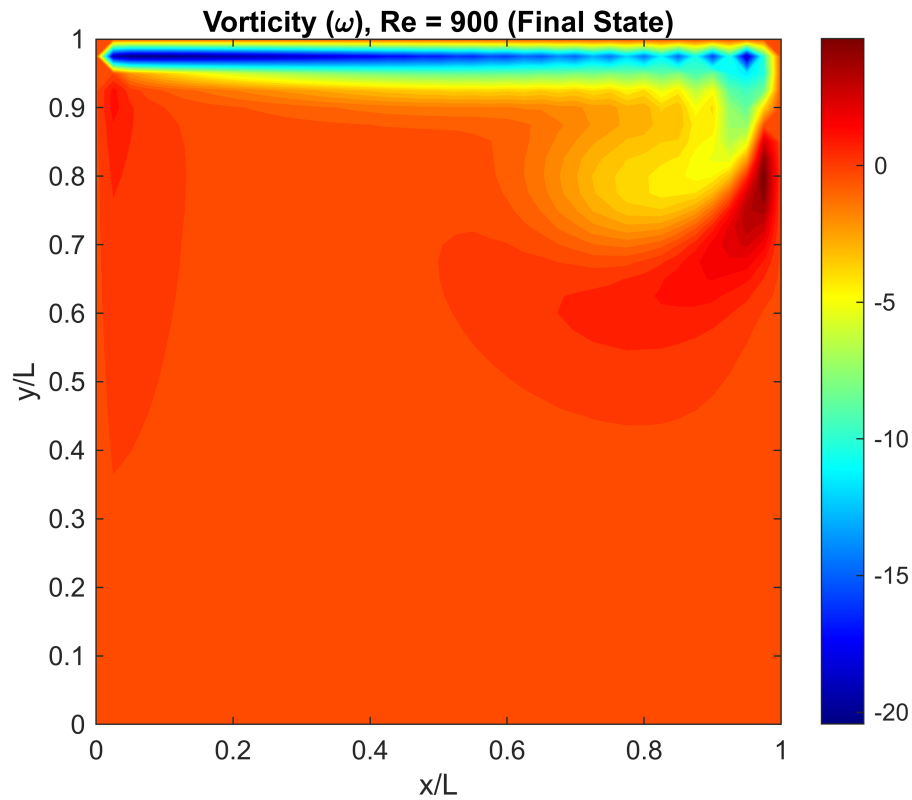
--- Finished Simulation for Re = 800 ---
--- Starting Simulation for Re = 900 ---
Re=900, Iteration 500/15000, Max Vel Change = 6.250036e-04
Re=900, Iteration 1000/15000, Max Vel Change = 3.982086e-04
Re=900, Iteration 1500/15000, Max Vel Change = 2.784552e-04
Re=900, Iteration 2000/15000, Max Vel Change = 1.680492e-04
Re=900, Iteration 2500/15000, Max Vel Change = 1.159145e-04
Re=900, Iteration 3000/15000, Max Vel Change = 8.950846e-05
Re=900, Iteration 3500/15000, Max Vel Change = 6.771585e-05
Re=900, Iteration 4000/15000, Max Vel Change = 5.305697e-05
Re=900, Iteration 4500/15000, Max Vel Change = 4.353432e-05
Re=900, Iteration 5000/15000, Max Vel Change = 3.677768e-05
Re=900, Iteration 5500/15000, Max Vel Change = 3.096376e-05
Re=900, Iteration 6000/15000, Max Vel Change = 2.848509e-05
Re=900, Iteration 6500/15000, Max Vel Change = 2.584618e-05
Re=900, Iteration 7000/15000, Max Vel Change = 2.351684e-05
Re=900, Iteration 7500/15000, Max Vel Change = 2.200906e-05
Re=900, Iteration 8000/15000, Max Vel Change = 2.040853e-05
Re=900, Iteration 8500/15000, Max Vel Change = 1.882090e-05
Re=900, Iteration 9000/15000, Max Vel Change = 1.727550e-05
Re=900, Iteration 9500/15000, Max Vel Change = 1.610787e-05
Re=900, Iteration 10000/15000, Max Vel Change = 1.527457e-05
Re=900, Iteration 10500/15000, Max Vel Change = 1.438192e-05
Re=900, Iteration 11000/15000, Max Vel Change = 1.340960e-05
Re=900, Iteration 11500/15000, Max Vel Change = 1.245527e-05
Re=900, Iteration 12000/15000, Max Vel Change = 1.160625e-05
Re=900, Iteration 12500/15000, Max Vel Change = 1.076986e-05
Re=900, Iteration 13000/15000, Max Vel Change = 1.007055e-05
Re=900, Iteration 13500/15000, Max Vel Change = 9.364512e-06
Re=900, Iteration 14000/15000, Max Vel Change = 8.737185e-06
Re=900, Iteration 14500/15000, Max Vel Change = 8.150291e-06
Re=900, Iteration 15000/15000, Max Vel Change = 7.568576e-06

```



Simulation time for Re=900: 266.56 seconds

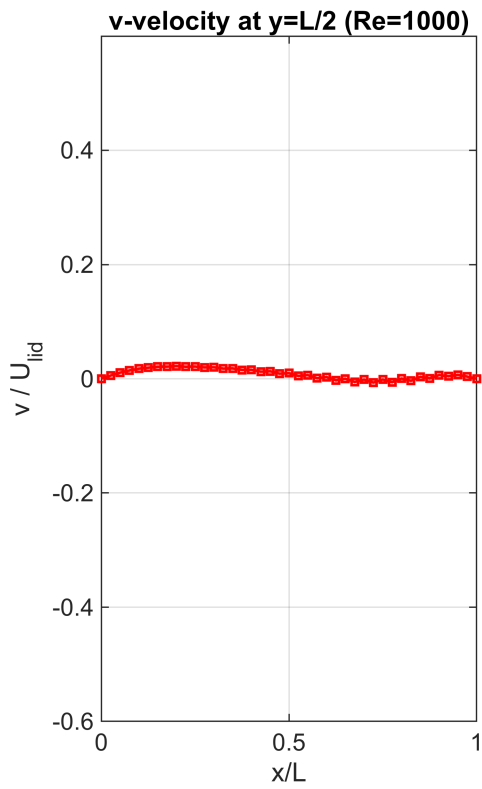
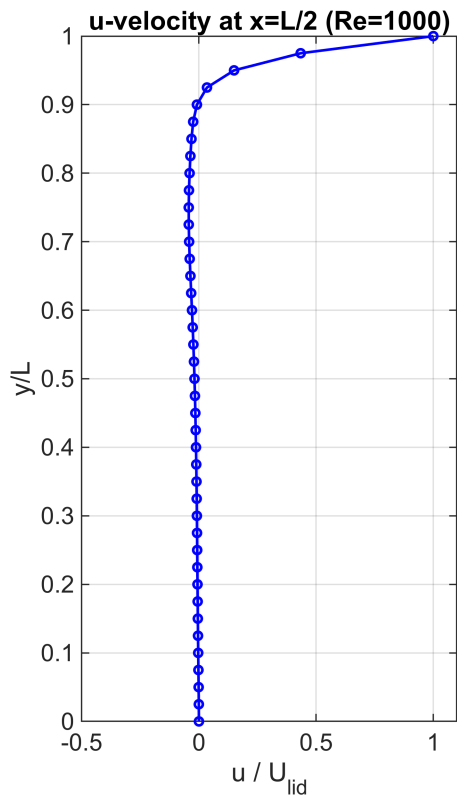
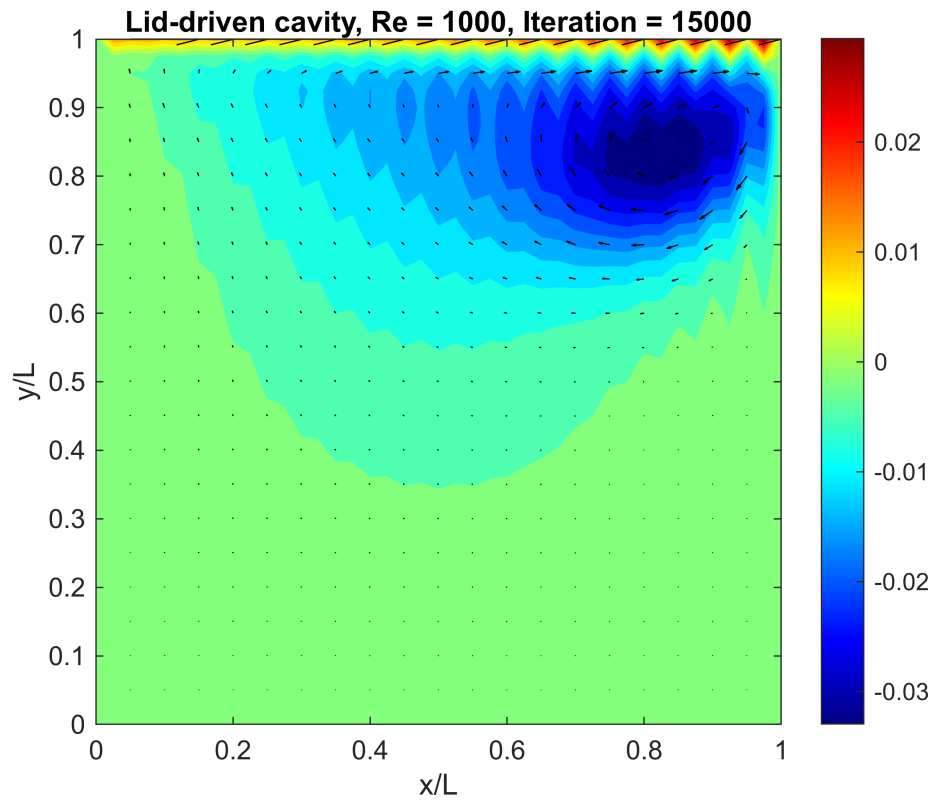
Warning: Solution for Re=900 did not fully converge within 15000 iterations. Final Max Vel Change = 7.568576e-06



```

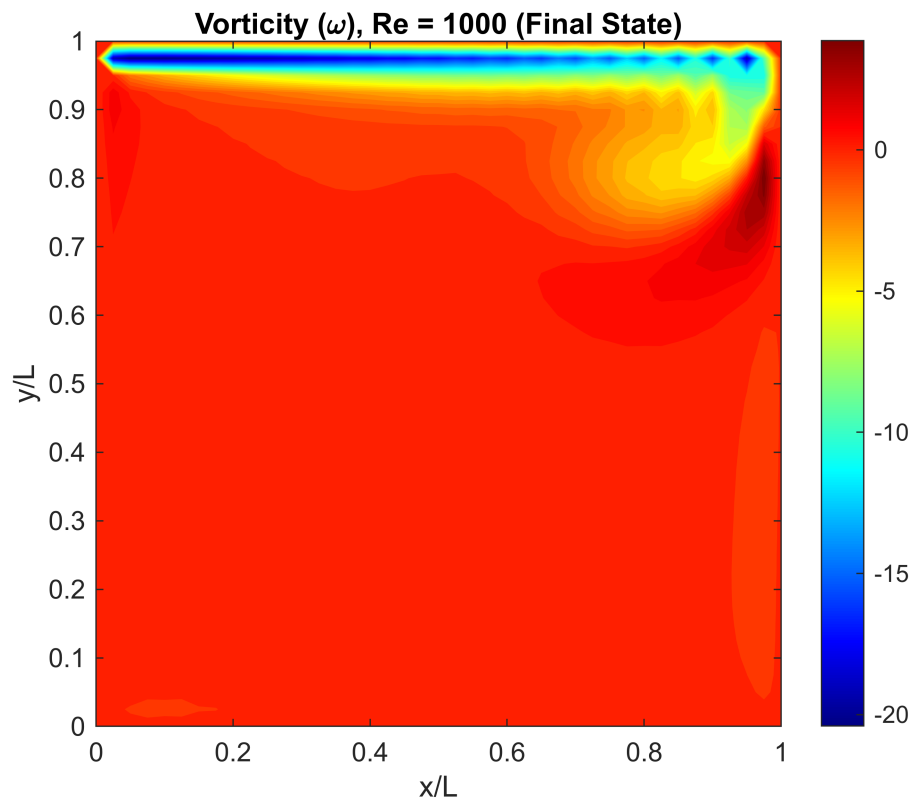
--- Finished Simulation for Re = 900 ---
--- Starting Simulation for Re = 1000 ---
Re=1000, Iteration 500/15000, Max Vel Change = 6.280759e-04
Re=1000, Iteration 1000/15000, Max Vel Change = 4.096476e-04
Re=1000, Iteration 1500/15000, Max Vel Change = 3.061013e-04
Re=1000, Iteration 2000/15000, Max Vel Change = 1.782630e-04
Re=1000, Iteration 2500/15000, Max Vel Change = 1.130904e-04
Re=1000, Iteration 3000/15000, Max Vel Change = 8.864666e-05
Re=1000, Iteration 3500/15000, Max Vel Change = 6.578171e-05
Re=1000, Iteration 4000/15000, Max Vel Change = 5.263944e-05
Re=1000, Iteration 4500/15000, Max Vel Change = 4.185259e-05
Re=1000, Iteration 5000/15000, Max Vel Change = 3.421203e-05
Re=1000, Iteration 5500/15000, Max Vel Change = 2.900757e-05
Re=1000, Iteration 6000/15000, Max Vel Change = 2.516971e-05
Re=1000, Iteration 6500/15000, Max Vel Change = 2.239679e-05
Re=1000, Iteration 7000/15000, Max Vel Change = 2.082650e-05
Re=1000, Iteration 7500/15000, Max Vel Change = 1.920958e-05
Re=1000, Iteration 8000/15000, Max Vel Change = 1.783477e-05
Re=1000, Iteration 8500/15000, Max Vel Change = 1.631867e-05
Re=1000, Iteration 9000/15000, Max Vel Change = 1.492071e-05
Re=1000, Iteration 9500/15000, Max Vel Change = 1.400721e-05
Re=1000, Iteration 10000/15000, Max Vel Change = 1.310877e-05
Re=1000, Iteration 10500/15000, Max Vel Change = 1.216456e-05
Re=1000, Iteration 11000/15000, Max Vel Change = 1.140693e-05
Re=1000, Iteration 11500/15000, Max Vel Change = 1.062590e-05
Re=1000, Iteration 12000/15000, Max Vel Change = 9.832778e-06
Re=1000, Iteration 12500/15000, Max Vel Change = 9.048978e-06
Re=1000, Iteration 13000/15000, Max Vel Change = 8.329750e-06
Re=1000, Iteration 13500/15000, Max Vel Change = 7.690193e-06
Re=1000, Iteration 14000/15000, Max Vel Change = 7.068557e-06
Re=1000, Iteration 14500/15000, Max Vel Change = 6.535350e-06
Re=1000, Iteration 15000/15000, Max Vel Change = 6.017771e-06

```



Simulation time for Re=1000: 262.56 seconds

Warning: Solution for Re=1000 did not fully converge within 15000 iterations. Final Max Vel Change = $6.017771e-06$



--- Finished Simulation for Re = 1000 ---

```
fprintf('\nAll simulations completed for Re = %s.\n', num2str(Re_values));
```

All simulations completed for Re = 100 200 300 400 500 600 700 800 900 1000.